Lecture 7 – Solar Photovoltaics Part I - Physics/Materials

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ECE180J

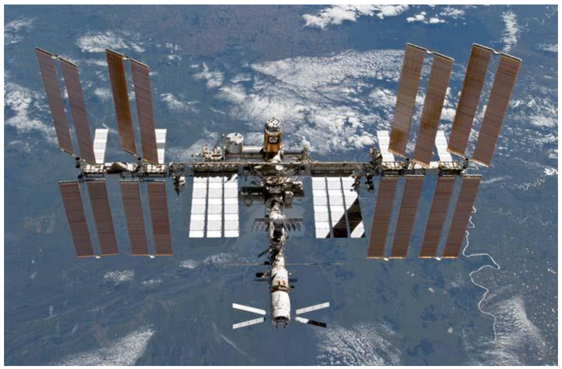
Outline

- PV Applications and History
- Conductor, Insulators, and Semiconductor
- Silicon: Doping and P/N-type
- Photon Energy and Band Gap
- PV Materials and Technologies

B3 Chap 4.1 – 4.6; B1 Chap 5
ECE 145 *Properties of Materials*
ECE 171 *Analog Electronics*

PV – Generation of Electricity from Light

- The net solar power input to the Earth is more than 8000 times humanity’s current rate of use of fossil and nuclear fuels.
- Solar photovoltaic (PV) cells directly convert solar energy into electricity in a solid state device.



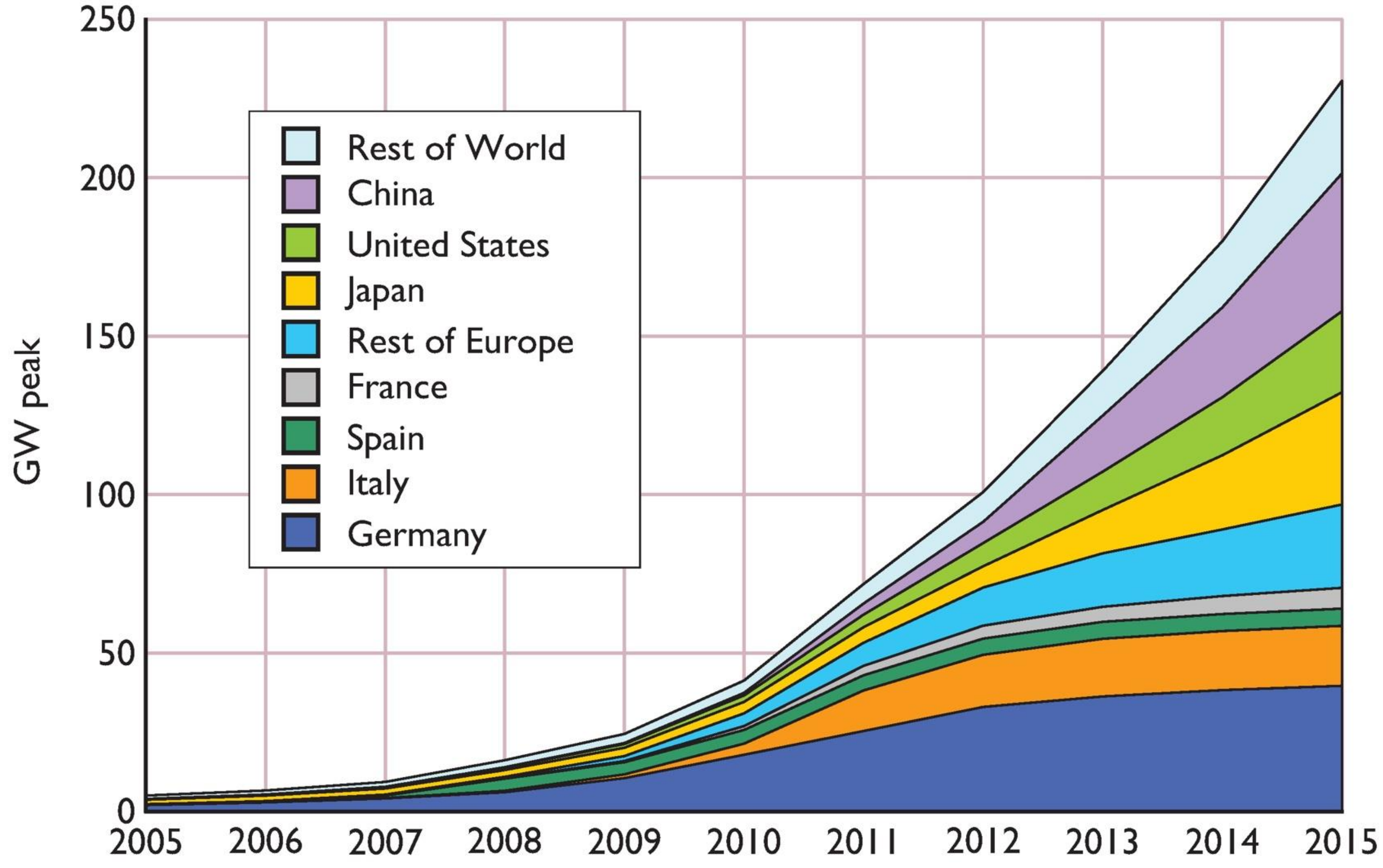
The Intl Space Station:
130 kW PV output

Solar car

Sunseeker Duo

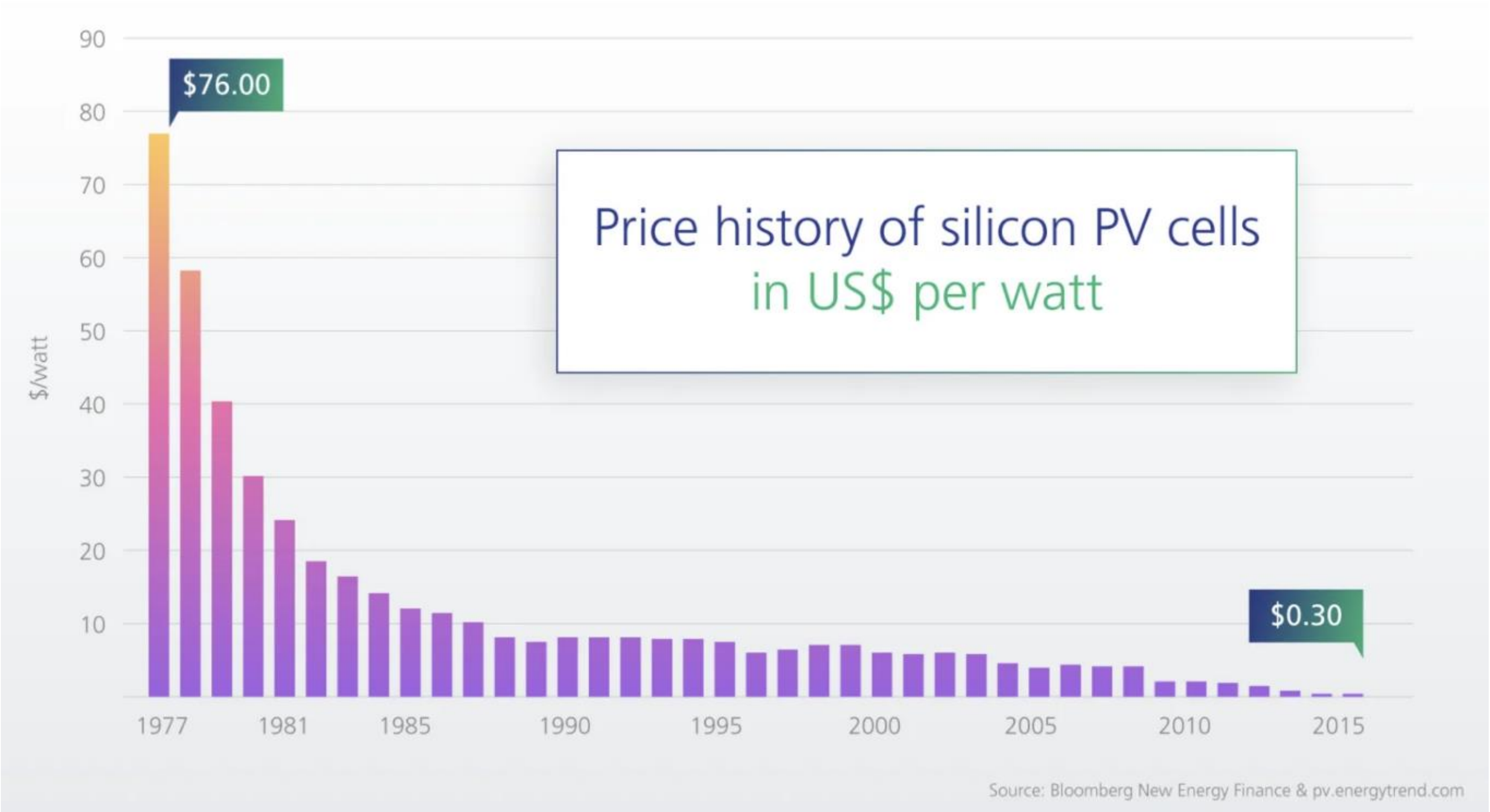
PlanetSolar

Development of PV (Peak Power)



Wp (Watt-peak):
Peak DC power delivered under idealized standard test conditions: Solar radiation of 1kW per square meter.

Development of PV: Price Drop



PV Module Costs

Learning rate: the percent drop in cost for each doubling of installed capacity.

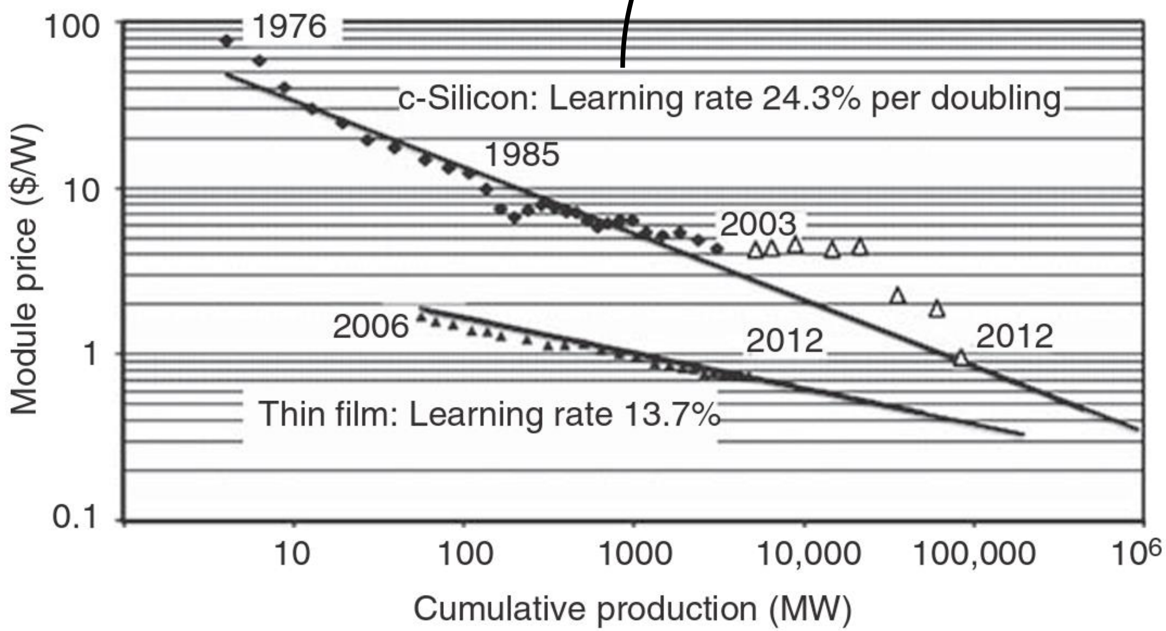
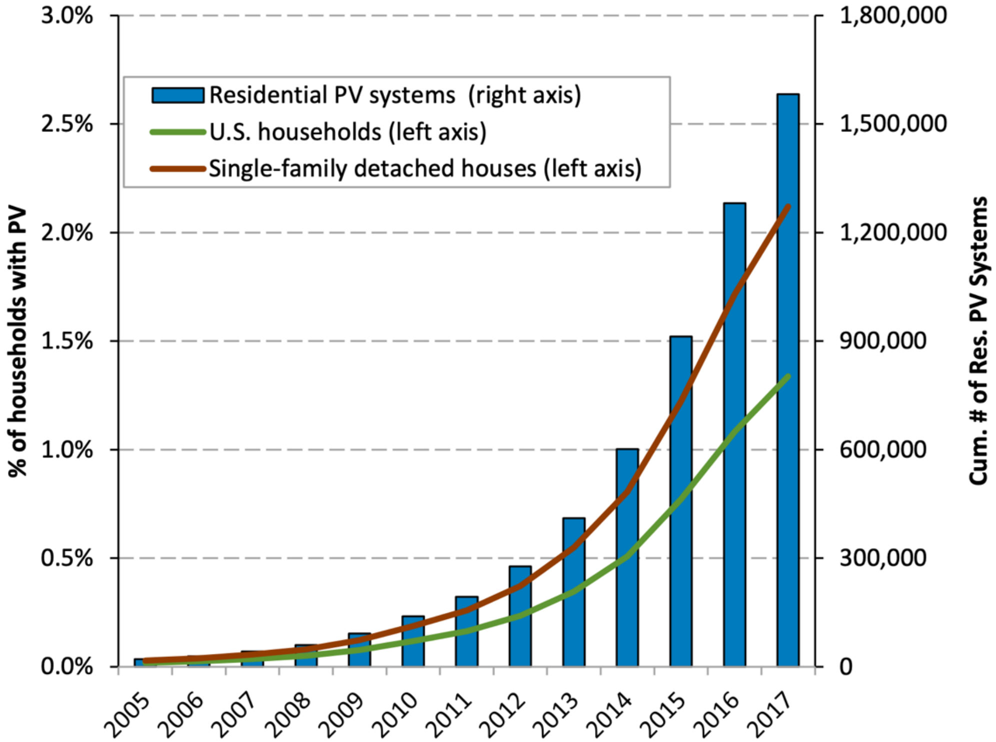


FIGURE 5.1 Photovoltaic module costs and cumulative production for both crystal silicon (c-Si) and thin-film technologies (Based on NREL data, 2012).

- Silicon cells use crystalline silicon wafers, while thin-film cells use very thin layers of materials such as CdTe, CIGS, or amorphous silicon deposited on a substrate.
- DoE's goal: By 2020, installed costs of $1/Wp for utility scale systems, $1.25/Wp for commercial rooftop PV, and $1.50/Wp for residential rooftop.

US Residential PV Penetration



Sources: Res. PV Installations: 2000-2009, IREC 2010 Solar Market Trends Report; 2010-2017, SEIA/GTM Solar Market Insight 2017 Year-in-Review; U.S. Households U.S. Census Bureau, 2015 American Housing Survey; state percentages based on 2000 survey.

- Since 2005, when the investment tax credit was passed by congress, the residential PV market has grown by approximately 44% per year, or about 81X.
- At the end of 2017, there were approximately 1.6 million residential PV systems in the United States.
- Still, only 1.3% of households own or lease a PV system (or about 2.1% of households living in single-family detached structures).
- However, solar penetration varies by location. Hawaii, California, and Arizona have residential systems on an estimated 31%, 11%, and 9% of households living in single-family detached structures.

Source: NREL

CALIFORNIA BECOMES THE FIRST STATE TO MAKE SOLAR PANELS MANDATORY

The new standard will bring the state closer to meeting its climate goals.

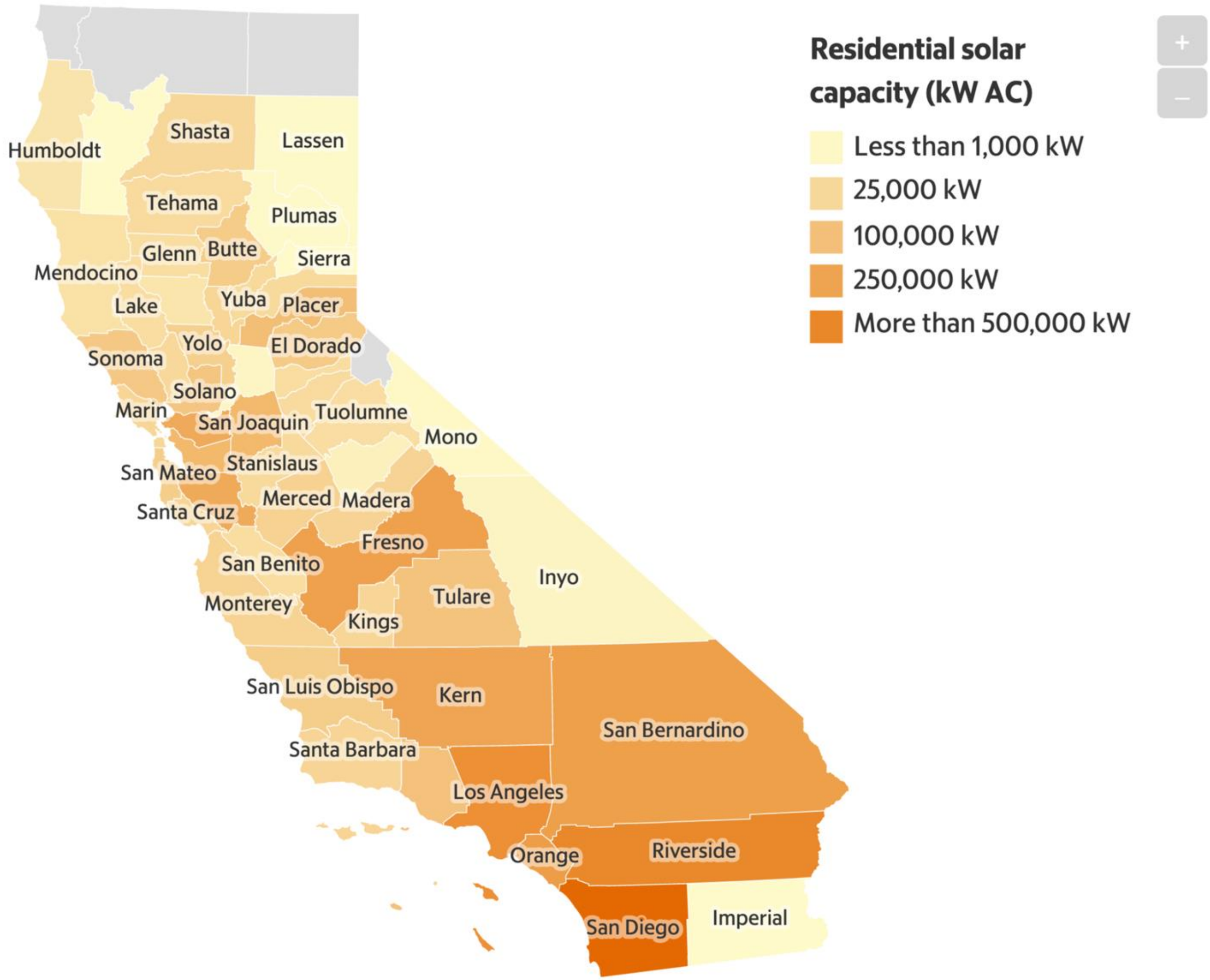
KATE WHEELING · MAY 9, 2018

Workers install solar panels on the roof of a home on May 9th, 2018, in San Francisco, California.
(Photo: Justin Sullivan/Getty Images)

Starting January 1st, 2020, virtually all new homes built in California will be equipped with solar panels.

- California law requires at least 50% of the state's electricity to come from noncarbon-producing sources by 2030.
- Solar power has increasingly become a driver in the growth of the state's alternative energy production.
- $10,000 additional upfront cost may be paid back by energy savings.

Residential Solar Capacity in CA



Figures reflect all net energy metering (NEM) interconnected solar photovoltaic (PV) systems in the Pacific Gas & Electric Co., Southern California Edison and San Diego Gas & Electric service territories as of June 30, 2019. No interconnected systems were reported in Alpine, Del Norte, Siskiyou or Modoc counties.

Map: Tim Sheehan | The Fresno Bee • Source: Go Solar Californnia • Get the data

Solar Canopy at UCSC

Home / 2021 / September / Solar array

Solar parking canopy goes online, providing UCSC with 2 megawatts of renewable energy

September 16, 2021

By Scott Hernandez-Jason

SHARE THIS STORY:

A two-megawatt solar parking canopy at the East Remote Parking Lot is now online, increasing the campus's use of renewable energy while reducing its energy bill.

The solar photovoltaic canopy will provide the campus with clean, reliable electricity for at least 20-years and save the campus an estimated $6 million on its energy bill. The array will generate enough energy to meet about 6% of our total campus electrical load.

"This is a major step towards directly reducing the campus' carbon footprint. As climate change impacts become more of a reality in all of our day-to-day lives, I am also encouraged that the chancellor is committed to exploring more options for future renewable energy projects on campus to keep us moving in the right direction," said Elida Erickson, director of the Office of Sustainability.

The project will generate more than three million kilowatt-hours (kWh) of electricity each year, enough to power more than 330 houses for a year, according to EPA calculations.

The project will generate more than three million kilowatt-hours (kWh) of electricity each year, enough to power more than 330 houses for a year, according to EPA calculations. (Photo by Nick Gonzales)

Solar Cell

A conventional crystalline silicon solar cell (as of 2005). Electrical contacts made from busbars (the larger silver-colored strips) and fingers (the smaller ones) are printed on the silicon wafer.

- PV: A material or device that can convert the energy contained in photons of light into an electrical voltage and current.
- A photon with short enough wavelength and high enough energy can cause an electron in a PV material to break free of the atom that holds it.
- If a nearby electric field is provided, those electrons can be swept toward a metallic contact where they can emerge as current.

Photovoltaic (PV) Effect

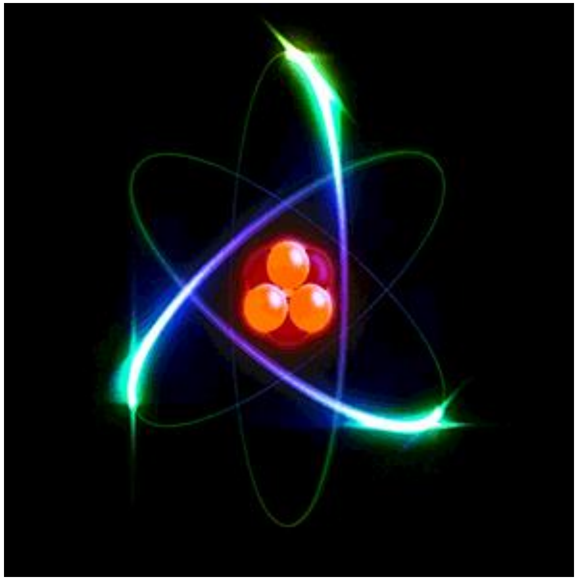
Photovoltaic effect: the generation of voltage and electric current in a material upon exposure to light.

- 1839 - Edmond Becquerel observed the **PV effect** via an electrode in a conductive solution exposed to light.
- 1883 - Charles Fritts developed a solar cell using selenium (<1% efficiency).
- 1904 - Wilhelm Hallwachs made a semiconductor-junction solar cell (copper & copper oxide).
- 1954 - Bell Labs announced the invention of the first practical silicon solar cell.

Edmond Becquerel (1820 – 1891), French physicist

Calvin S. Fuller at work diffusing boron into silicon to create the world's first solar cell

Atoms



An atom consists of a nucleus of protons and neutrons, surrounded by an electromagnetically bound swarm of electrons.

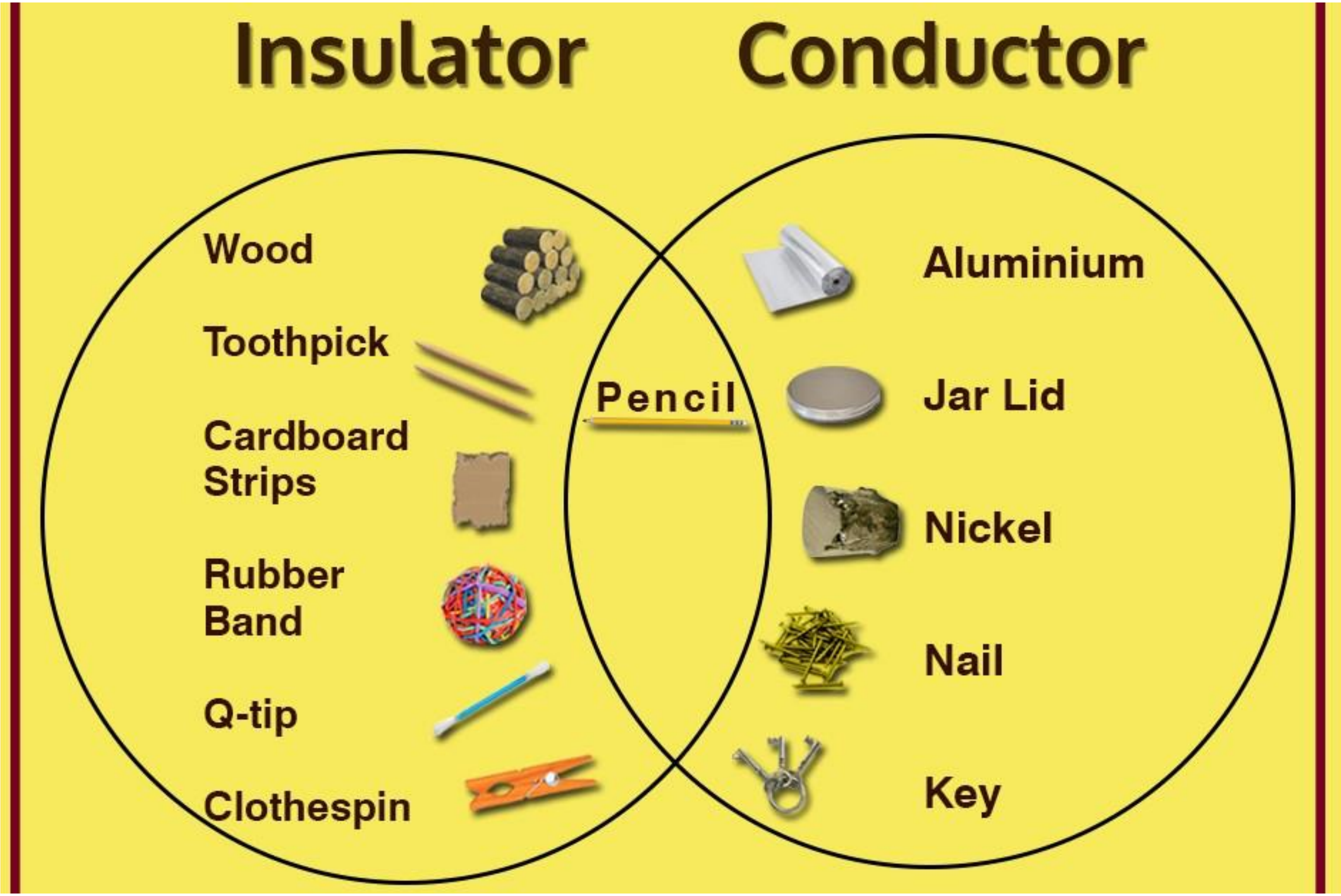
- All matter is made up of atoms, which consist of

- Protons (positively charged)
- Neutrons (no net electric charge).
- Electrons (negatively charged) are moving in circular orbits at specific energy levels around the nucleus (by Bohr model).

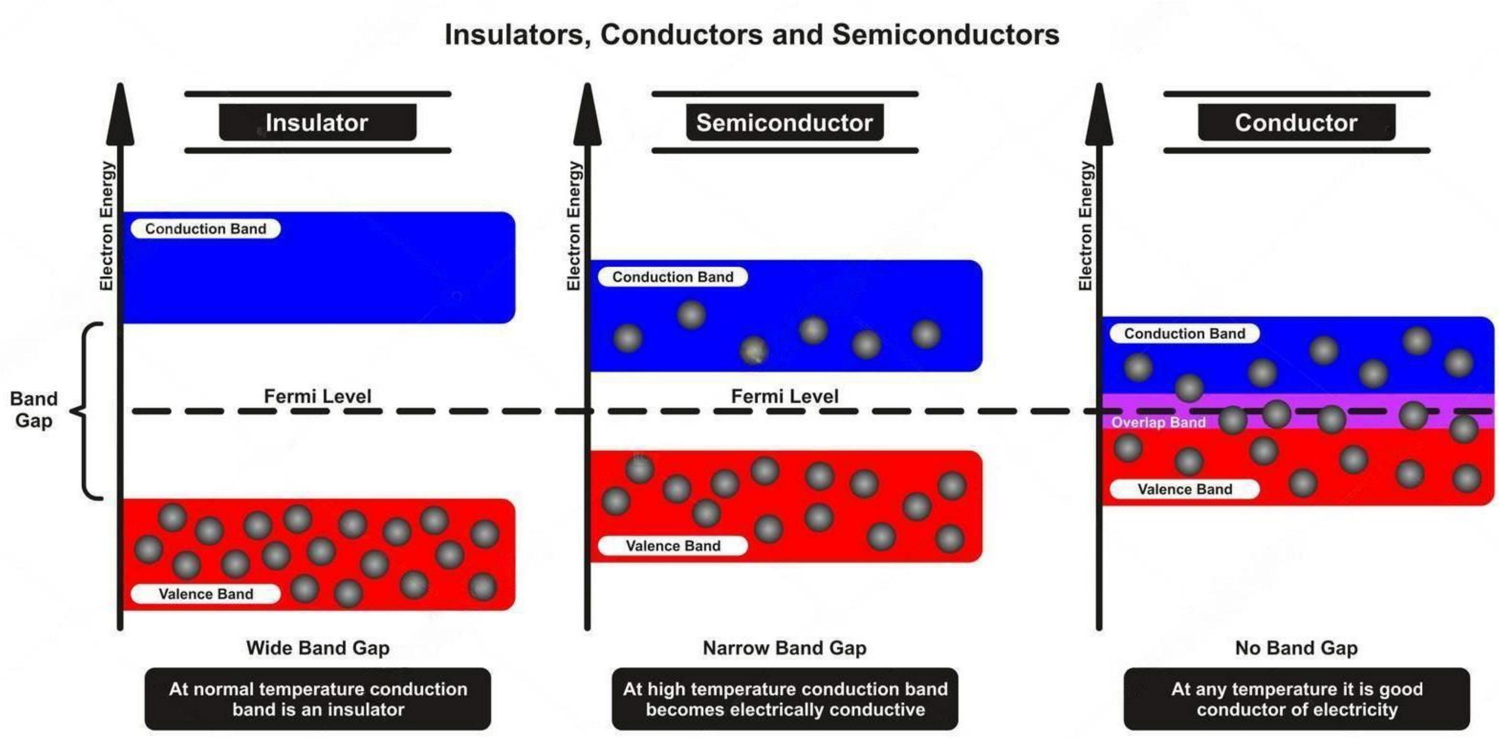
- Protons and neutrons each have masses close to 1 amu ($\approx 1.66 \times 10^{-27}$ kg), while electrons are much lighter ($\approx$1/1836 amu). Thus, more than 99.9% of an atom's mass is concentrated in its nucleus.

- Electrons determine chemical properties, contribute to electricity and light emission, while protons define atomic identity and stability, and neutrons stabilize the nucleus and influence isotopic properties.

Conductors & Insulators

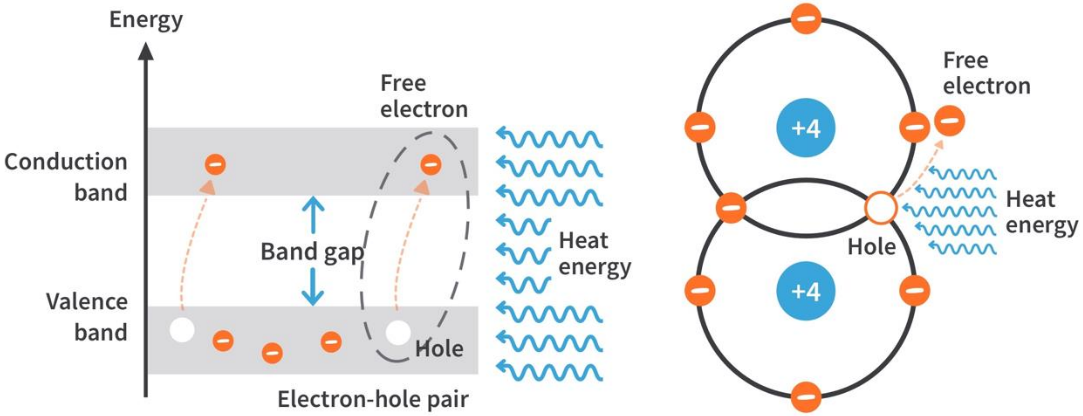


Band Gap of Semiconductors

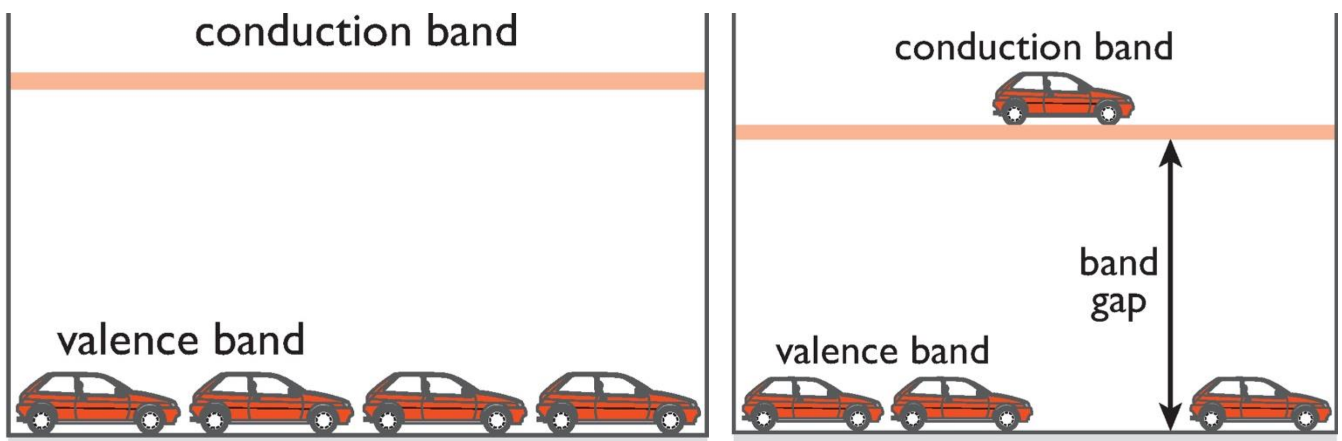


- Fermi Level is the maximum energy point that an electron could reach at zero Kelvin.
- The valence and conduction bands are the bands closest to the Fermi level and determine the electrical conductivity.
- The closer the Fermi level is to the conduction band, the easier electron transitions from the valence to conduction band become.

Electron Excitation



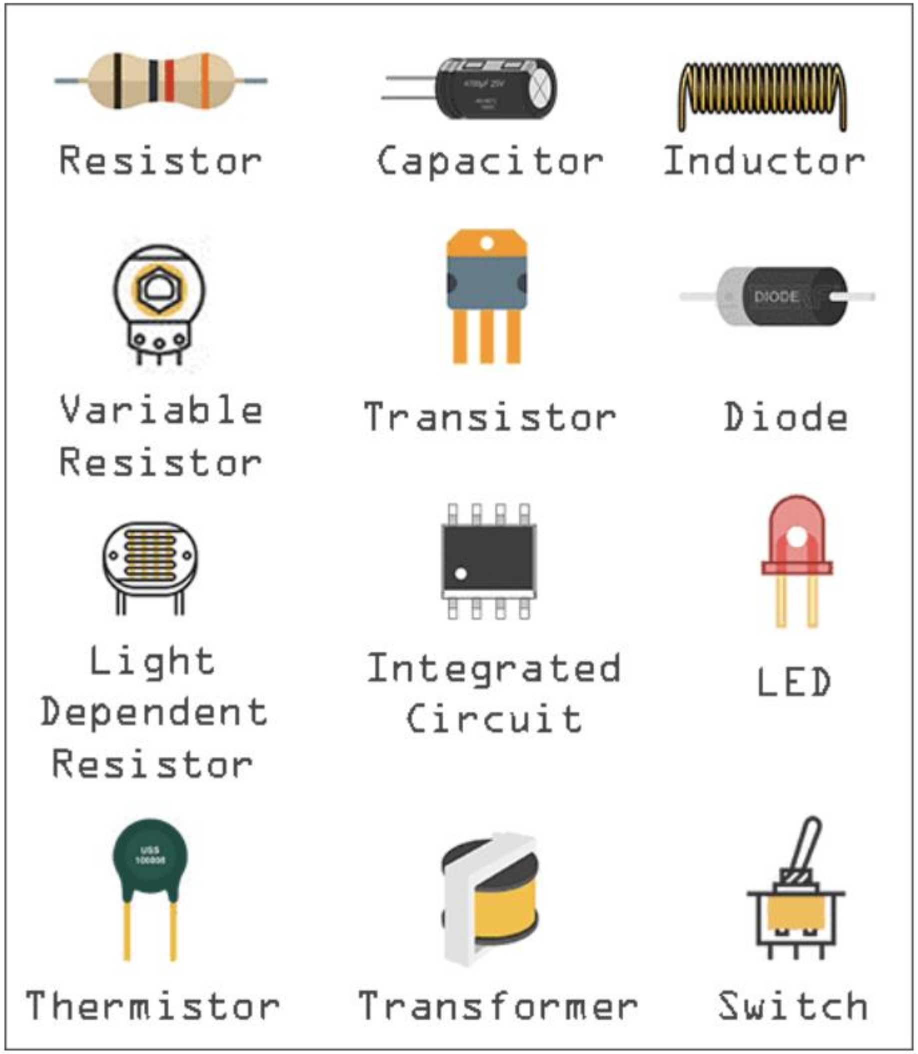
Electron excitation is when electrons gain energy, moving from lower to higher energy states, often by absorbing energy from sources like light, heat, or electrical fields.



‘Car-parking’ analogy

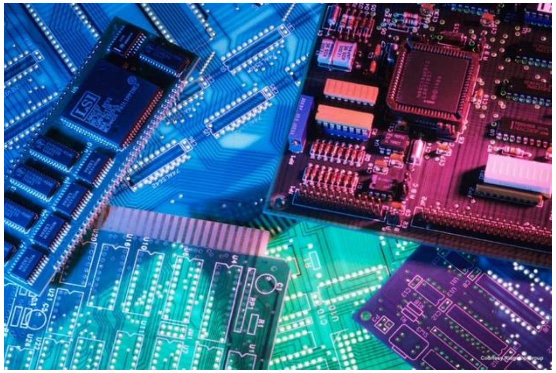
Left: 2nd floor is empty, but the 1st floor is full, no car can move.
Right: a car is ‘promoted’ to the 2nd floor, and can then move around freely.
This leaves a ‘hole’ that allows 1st floor cars to move around.

Semiconductor Devices



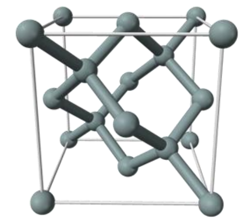
Semiconductors have a small band gap:

- Allows for a fraction of the valence electrons of the material to move into the conduction band given a certain amount of energy.
- Allows semiconductors to convert light into electricity in PV cells.

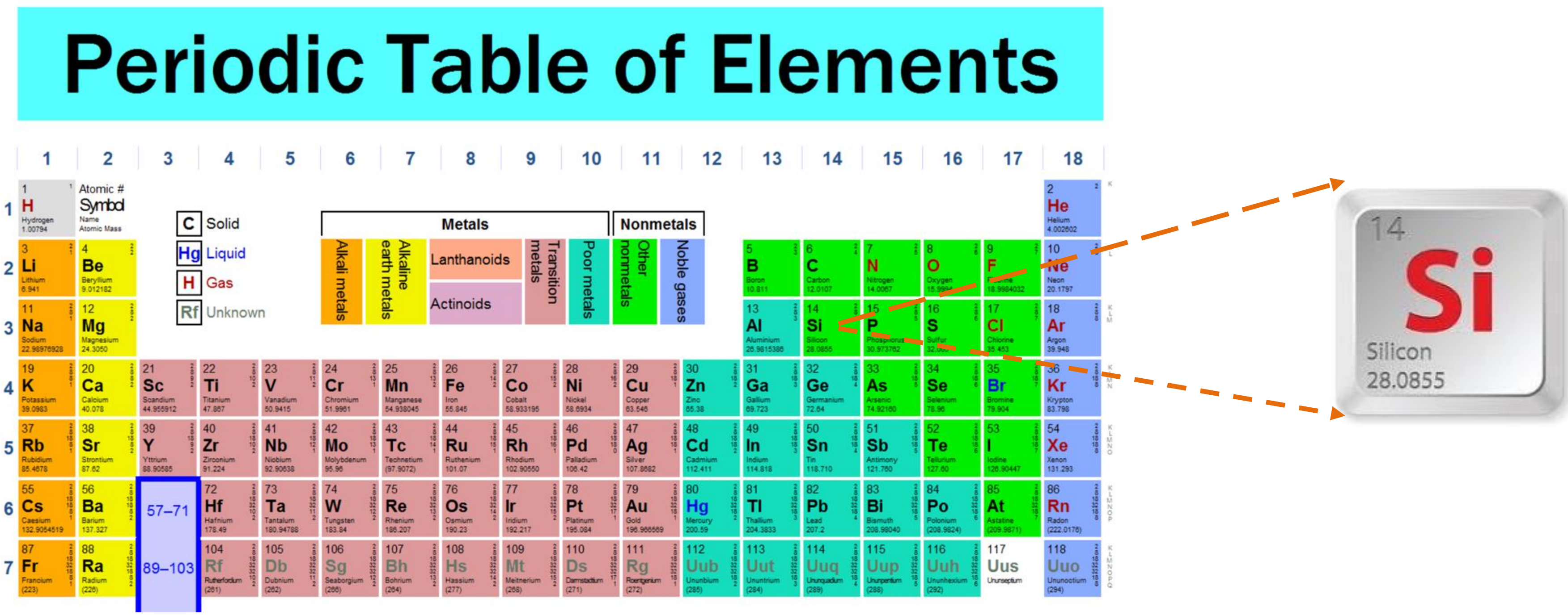


The Workhorse: Silicon

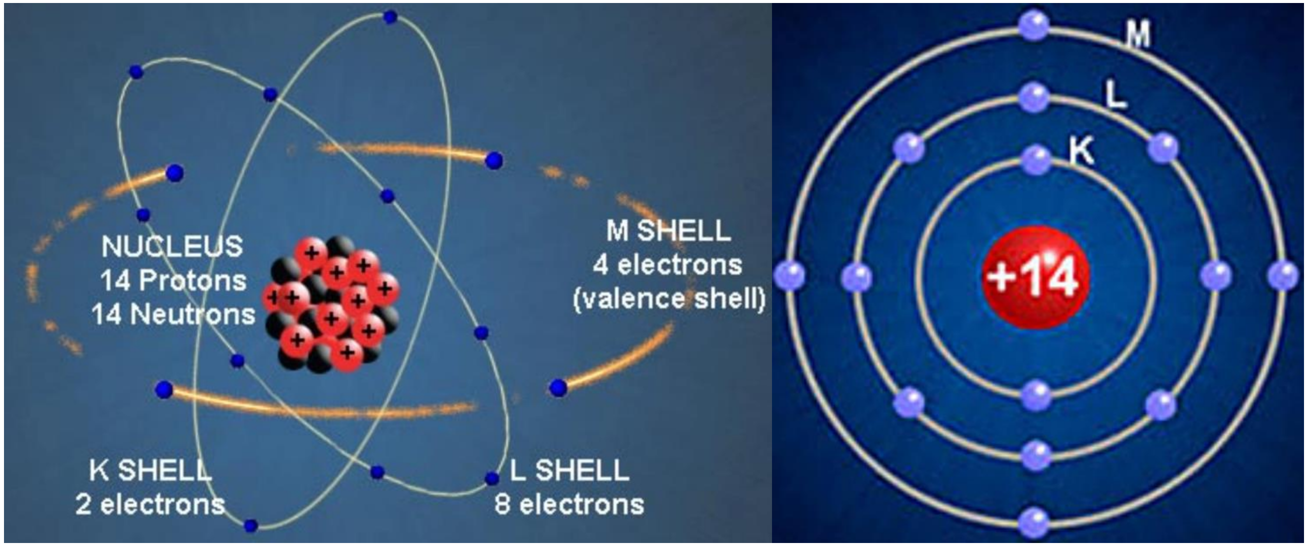
- PV cell often is made almost entirely from silicon, the second most abundant element in the Earth's crust.
- No moving parts and can operate for an indefinite period without wearing out.



Jöns Jacob Berzelius, discoverer of silicon



Silicon – Atomic Structure



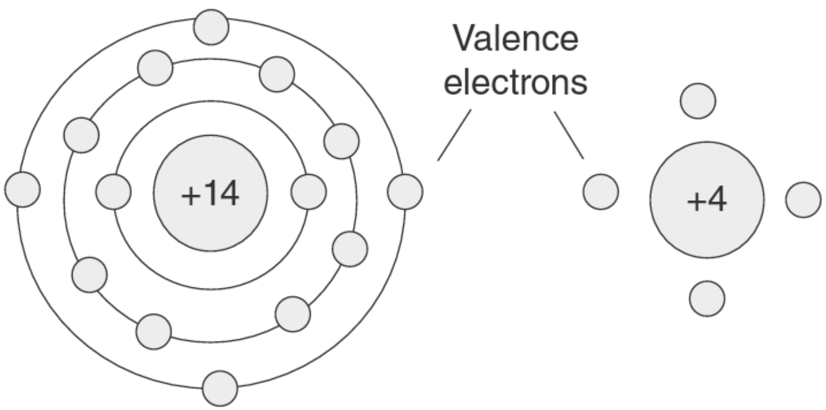
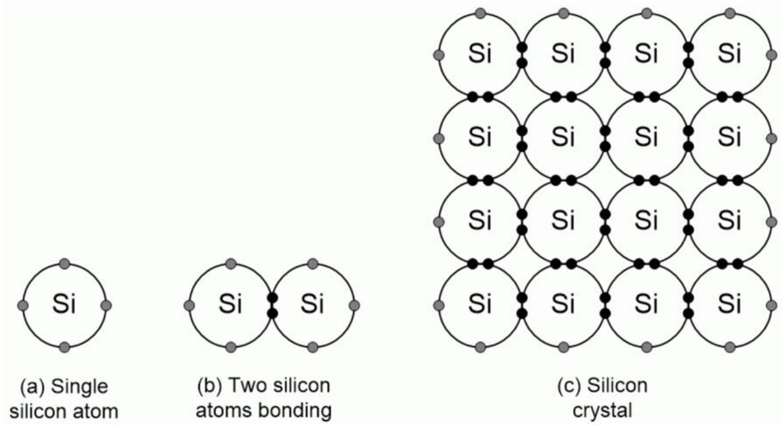


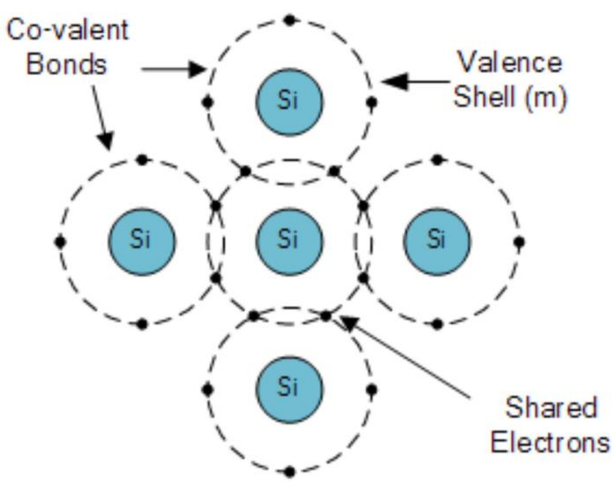
Figure: (a) Pictorial view; (b) schematic view; (c) simplified view, where 4 outer valence electrons are spinning around a nucleus with a +4 charge.

Silicon Crystal



Silicon atoms bonding (Source: Max Maxfield)

Consider a solid (like a semi-conductor) as any material in which the atoms are arranged in an ordered fashion (referred to as crystal or lattice).



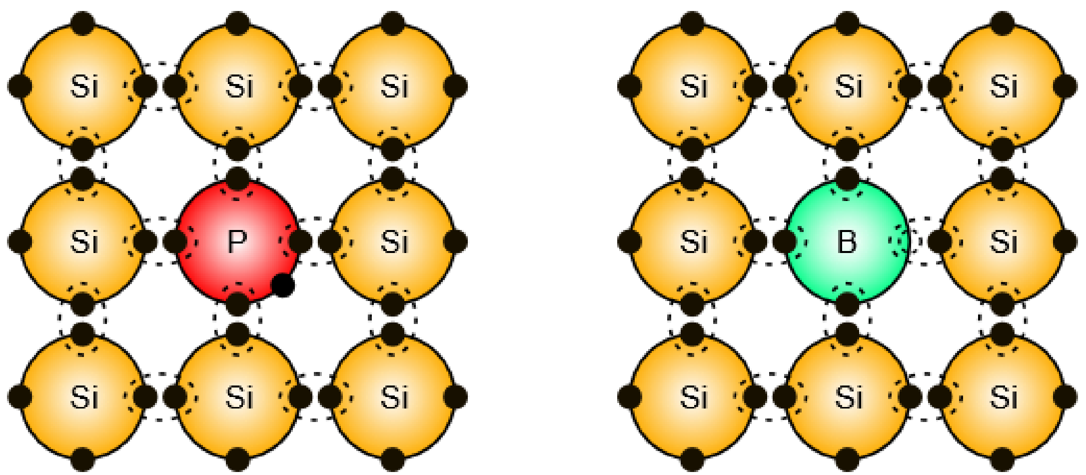
Silicon Crystal Lattice

- Only the most outer 4 valence electrons can help bond with other atoms.
- In solid silicon, each atom forms covalent bonds with four neighboring atoms, sharing one valence electron with each.
- This arrangement gives the illusion that each atom has a full outer shell with eight electrons.

Source: http://homework.uoregon.edu/pub/class/155/Canvas/semi.html

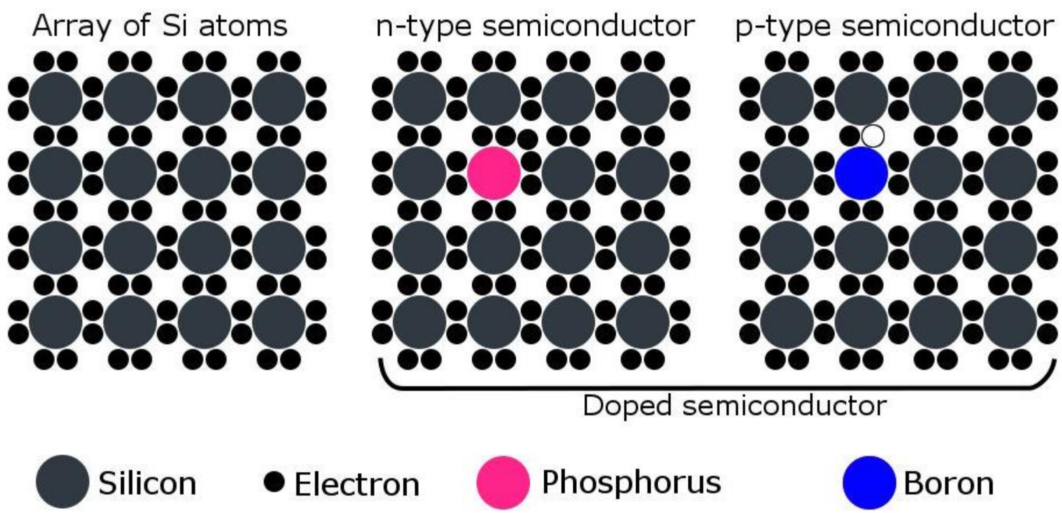
Doping

- Impurities and temperature can affect the Fermi level.
- Semiconductor atoms are closely grouped together in a crystal lattice and so they have very few free electrons to be good conductors.
- The ability of semiconductors to conduct electricity can be greatly improved by introducing donor or acceptor atoms to the crystalline structure, either producing more free electrons or more holes.
- This process is called “doping” and as the semiconductor material is no longer pure, these donor and acceptor atoms are collectively referred to as “impurities”.



n-type p-type

P-type & N-type

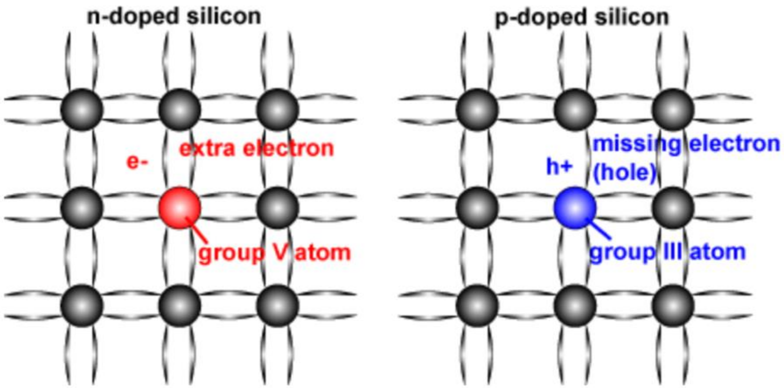


PV cells consist of a junction between thin layers of two different types of semiconductor:
p-type (positive)
n-type (negative)

- **N-type:** Doped with phosphorus such that the doped material possesses a surplus of free electrons.

- **P-type:** Doped with boron causing the material to have a deficit of free electrons. Missing electrons are called 'holes', which can be considered equivalent to a positively charged particle.

P-type & N-type Comparison



A silicon crystal lattice doped with impurities

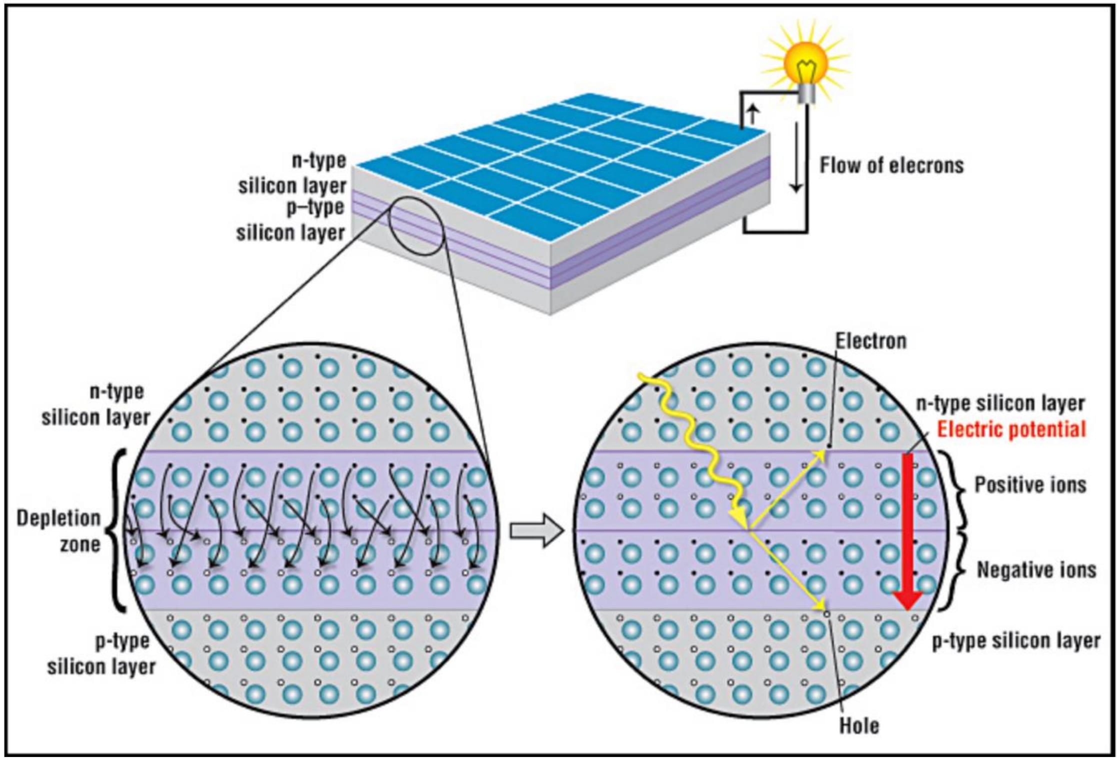
	N-type (negative)	P-type (positive)
Dopant	Group V (e.g. Phosphorous)	Group III (e.g. Boron)
Bonds	Excess Electrons	Missing Electrons (Holes)
Majority Carriers	Electrons	Hole
Minority Carriers	Holes	Electrons

having 5 valence electrons → Donors; having 3 valence electrons → Acceptors

Donors (column V elements) (add more electrons)		Acceptors (column III elements) (add more holes)	
P	Phosphorus (most common)	**B**	Boron (most common)
As	Arsenic	**Ga**	Gallium
Sb	Antimony	**In**	Indium
		Al	Aluminum

Source: https://www.pveducation.org/pvcdrom/pn-junctions/doping

P-N Junction: Depletion Zone



- When p- and n-type materials are brought together: Electrons **diffuse** from n → p (high → low concentration); and holes **diffuse** from p → n
- These carriers recombine near the junction, leaving behind: Positive donor ions (n-side); and negative acceptor ions (p-side)
- This creates: A depletion region and an internal electric field, which opposes further carrier diffusion.

Source: https://www.acs.org/

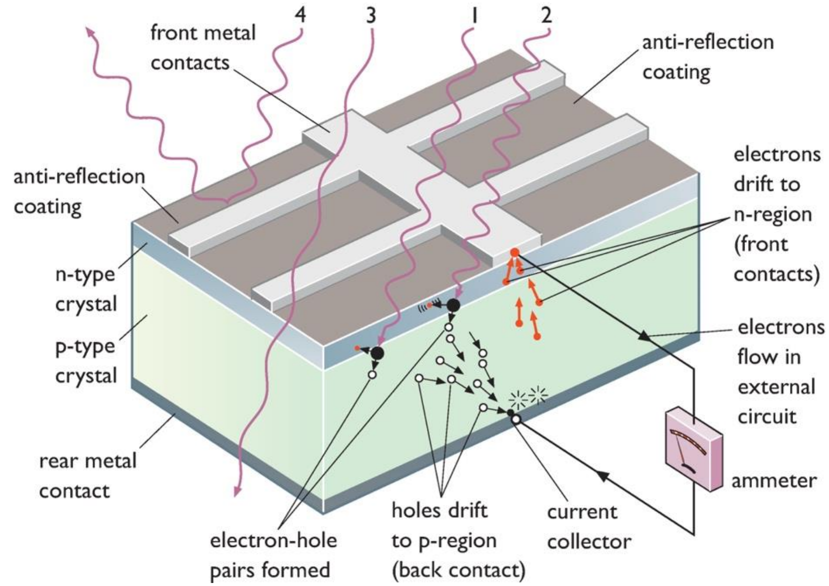
P-N Junction: Free Carriers vs Fixed Ions

Aspect	Fixed ions (lattice charges)	Free carriers (electrons & holes)
Origin	Ionized dopants (donors/acceptors)	Thermally or optically generated carriers
Charge type	p-side: **negative (acceptors)** n-side: **positive (donors)**	Electrons: negative Holes: positive
Mobility	**Immobile** (fixed in lattice)	**Mobile** (drift & diffusion)
Location	Mainly in **depletion region**	Mainly in **quasi-neutral regions**
Role in junction	Create **internal electric field**	Carry **current**
Response to electric field	Do **not move**	**Move (drift)** under field
Behavior at equilibrium	Remain fixed	**Drift current = diffusion current (opposite directions) → net current = 0**
Under illumination	Unchanged	Generated as **electron–hole pairs**
Contribution to current	None	**Primary contributors** to current

Source: https://www.acs.org/

Electron Excitation and Carrier Separation

- When sunlight is absorbed, electron–hole pairs are generated:
  1) Photon absorption: A photon with energy E strikes the semiconductor. If $E \geq E_g$ (bandgap energy), the photon can be absorbed.
  2) Electron excitation: The photon transfers its energy to an electron in the valence band, lifting it to the conduction band.
  3) Hole formation: The vacancy left behind in the valence band behaves as a hole (positive charge carrier).

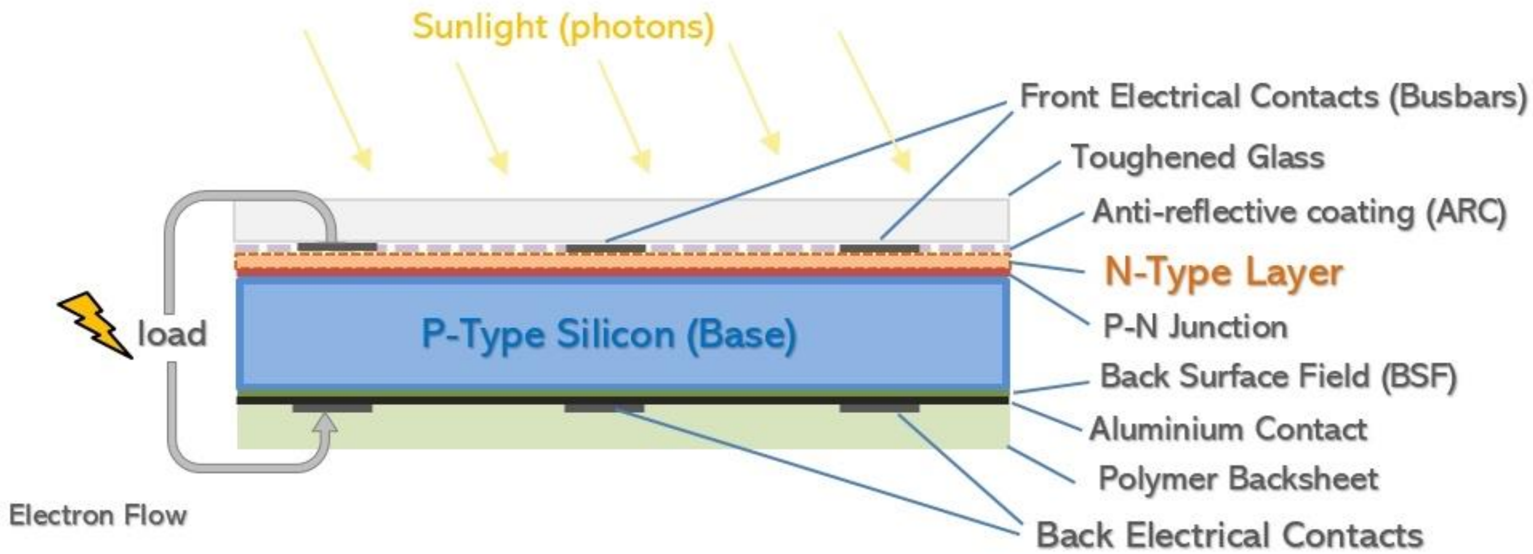


- If this occurs within or near the depletion region, the internal electric field separates carriers via **drift** process: electrons → n-side, holes → p-side.
- Electrons are collected at the front contact, flow through the external circuit (producing current), and return to the p-side, where they recombine with holes.

Diffusion vs. Drift in Semiconductors

Aspect	Diffusion	Drift
Driving force	**Concentration gradient**	**Electric field E**
Physical cause	Random thermal motion	Force qE on charges
Direction	High → low concentration	Along field (holes) / opposite field (electrons)
Requires electric field?	No	Yes
Current expression	$J \propto \frac{dn}{dx}, \frac{dp}{dx}$	$J \propto E$
Speed	Relatively slower	Faster with strong field
Dominant region (p–n junction)	Quasi-neutral regions	Depletion region
Role in PV cell	Brings carriers to junction	Separates and collects carriers
At equilibrium	Balanced by drift	Balanced by diffusion
Under illumination	Supplies carriers to junction	Drives carriers to contacts

Thickness and Placement of P/N Layers



- Layer thickness is designed based on mobility, lifetime, and diffusion length of minority carriers.
- Electrons have higher mobility and longer diffusion length than holes.
- The p-type base (electron minority) is made thick to enable efficient collection.
- The n-type emitter (hole minority) is made thin due to short diffusion length.
- The emitter is on top and thin so light can reach the p-type region.

Photon Energy

The energy of a photon of light: $\boxed{E = hf = hc/\lambda}$

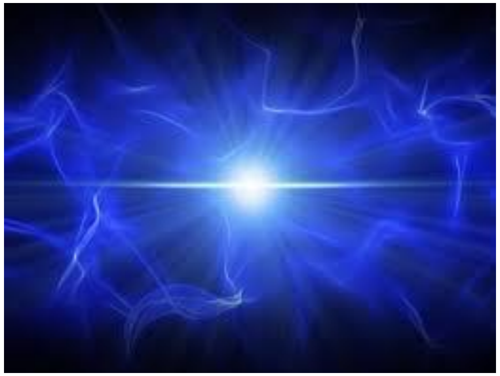
$$\text{Planck's constant: } h = 6.626 \times 10^{-34}\ J \cdot s$$

$$\text{light speed: } c = 3 \times 10^{8}\ m/s \qquad \text{light wavelength: } \lambda$$

Energy at the atomic level is normally described by electron-volts (eV).

1 eV is the energy that an electron acquires when its voltage is increased by 1 V.

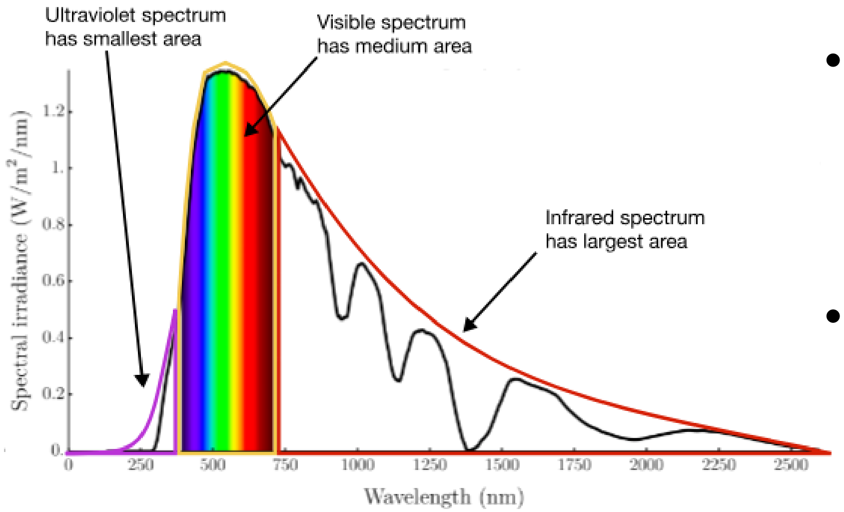
$$1\ eV = 1.6 \times 10^{-19}\ J \longrightarrow \boxed{E_{(eV)} = 1241.5/\lambda_{(nm)}}$$



A photon of blue light (450nm): 2.76eV

A photon of infrared light (900nm): 1.38eV

PV Cell Efficiency Limits

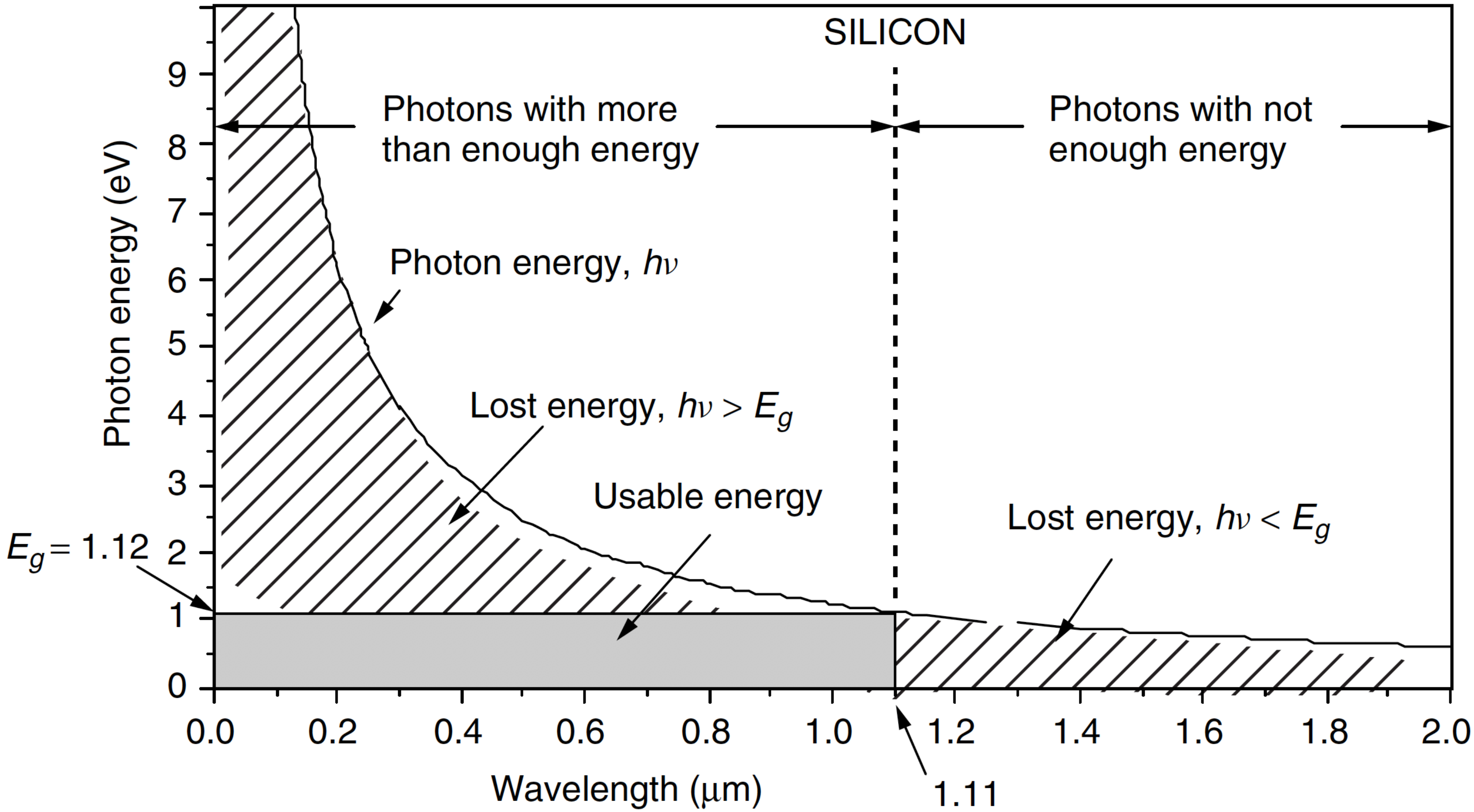


- The entire spectrum of sunlight, from infrared to ultraviolet, covers a range of about 0.5 eV to about 2.9 eV.
- Solar cells are not 100% efficient ← semiconductors do not respond to the entire spectrum of sunlight.

- Photons with energy less than silicon's bandgap are not absorbed, which wastes about 18% of incoming energy.
- The energy content of photons above the bandgap will be wasted surplus re-emitted as heat or light. This accounts for an additional loss of about 49%.
- Thus about 67% of energy from the original sunlight is lost; i.e., only 33% is usable for generating electricity in an ideal solar cell.

- The useful electrical energy that a photon contributes to the cells is only equal to the band gap.

Useful and Wasted Energy



- Photons with wavelengths above 1.11 μm don't have the 1.12 eV needed to excite an electron, and this energy is lost.
- Photons with shorter wavelengths have more than enough energy, but any energy above 1.12 eV is wasted.

Band Gaps of Different Materials

TABLE 5.2 Band-Gap and Cut-off Wavelength Above Which Electron Excitation Does Not Occur

Quantity	Si	a-Si	CdTe	$CuInSe_2$	$CuGaSe_2$	GaAs
Band-gap (eV)	1.12	1.7	1.49	1.04	1.67	1.43
Cut-off wavelength (μm)	1.11	0.73	0.83	1.19	0.74	0.87

- **Si** = crystalline silicon (ordered, high-efficiency)
- **a-Si** = amorphous silicon (disordered, lower-efficiency, flexible applications)

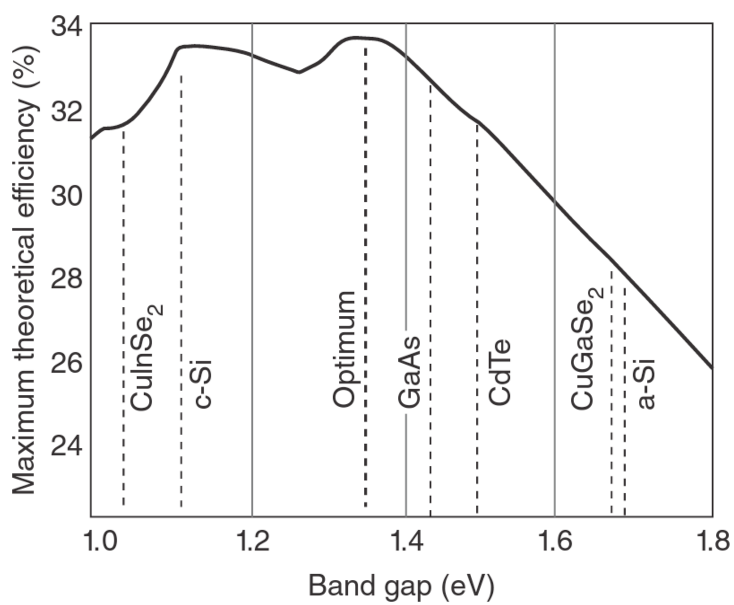


Fig: The Shockley-Queisser limit for solar cell efficiency (single-junction, unenhanced insolation)

Quiz

Q1. Blue light of wavelength 485 nm falls on a silicon photocell made from a semiconductor with bandgap is 1.35 eV. What is the maximum fraction of the light's energy that can be converted into electrical power?

Only the fraction of the incident energy equal to the bandgap energy can be converted to electrical power, the rest is lost as heat (the semiconductor hears up). Therefore, the efficiency is given as

$$\eta = \frac{E_g}{E_{485nm}} \times 100\% = \frac{1.35eV}{\frac{hc}{\lambda}} \times 100\% = \frac{1.35eV}{1241.5eV \cdot nm / 485nm} \times 100\% = 52.74\%$$

Q2. Find the maximum wavelength of the photon needed to cause an electron to jump from the valence to the conduction band in (a) silicon, band gap 1.1 eV; (b) cadmium selenide, band gap of 1.73 eV; (c) cadmium sulfide, band gap 2.42 eV.

Using $\lambda = \frac{hc}{E}$

(a) $\lambda = 1241.5eV \cdot nm / 1.1eV = 1128.64nm$

(b) $\lambda = 1241.5eV \cdot nm / 1.73eV = 717.63nm$

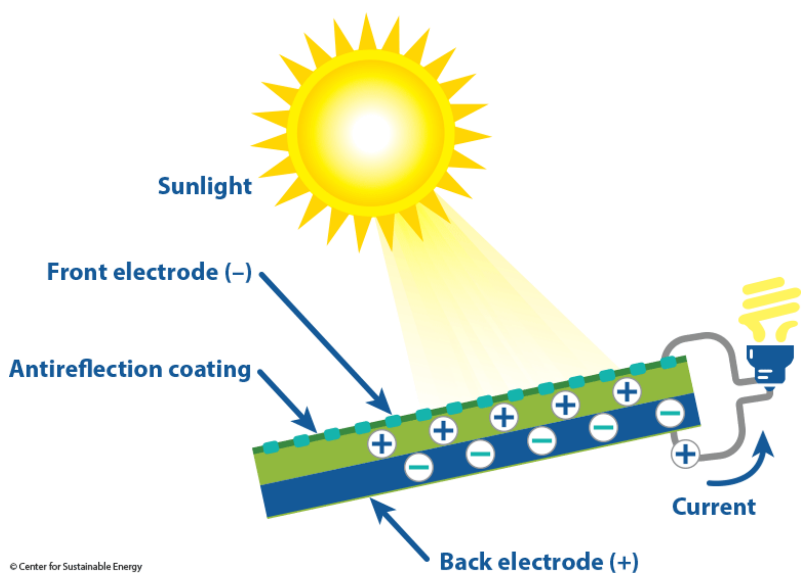
(c) $\lambda = 1241.5eV \cdot nm / 2.42eV = 513.02nm$

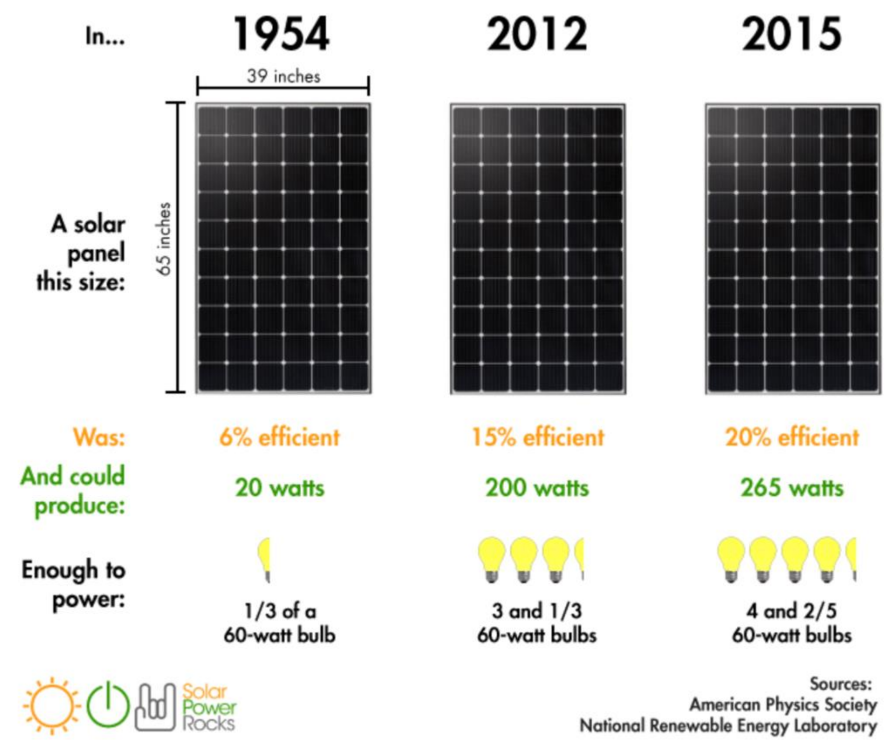
PV Cells & Modules

Q: How much voltage does a typical silicon PV cell produce?

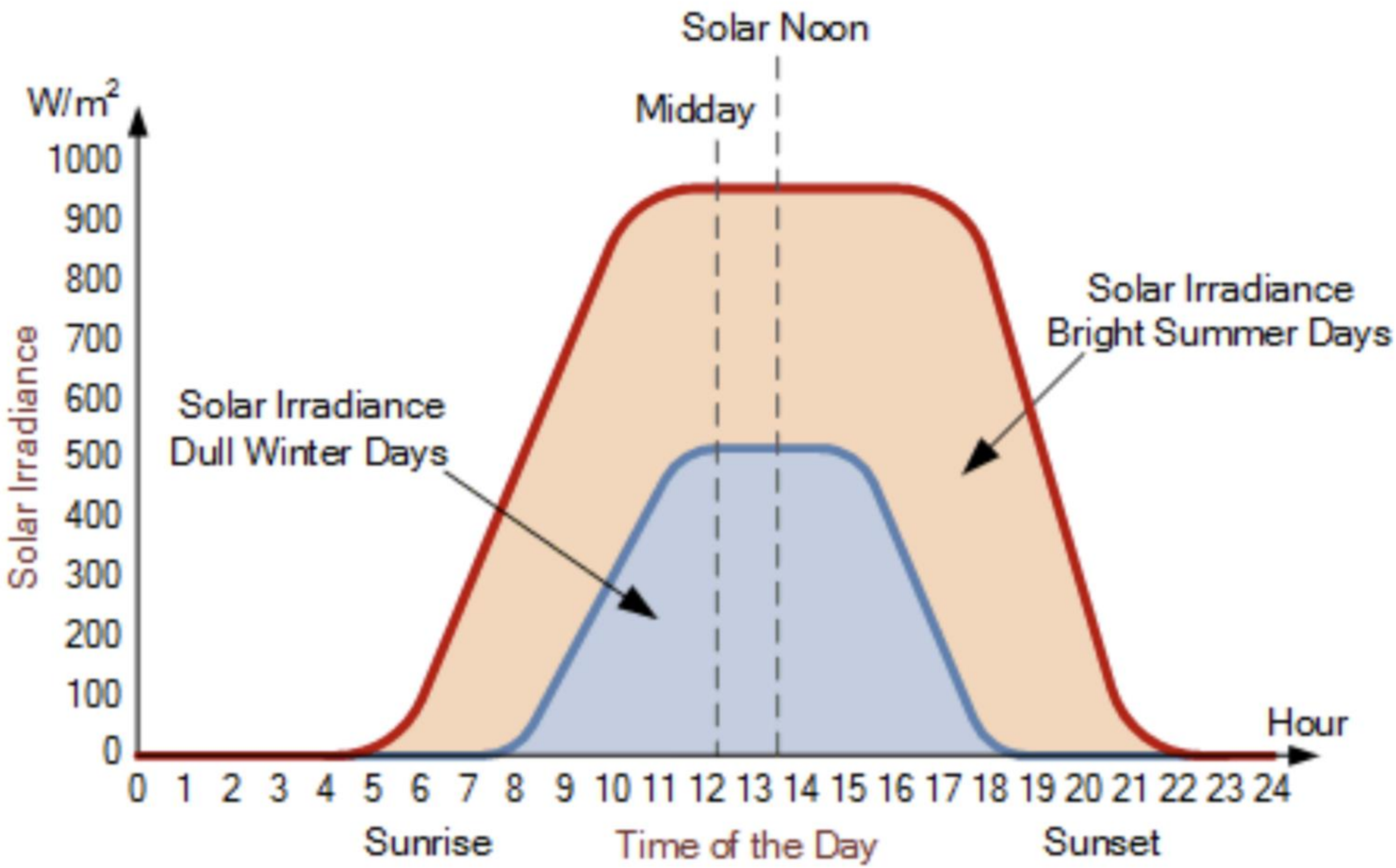
Answer:

- A typical silicon PV cell produces around 0.5 volts.
- 60-72 cells are combined (series-parallel mix) into PV modules.
- They are normally around 1.5 m^2 in area, and have a peak power output of 120-300 watts, depending on the design/tech.



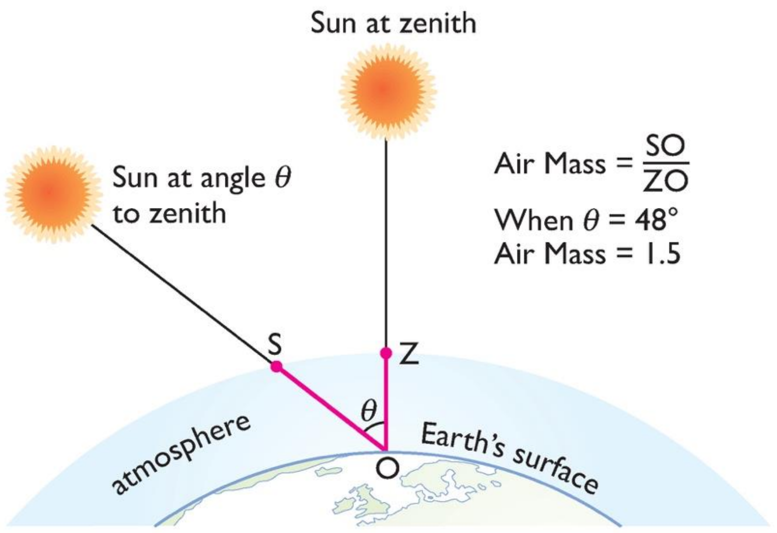


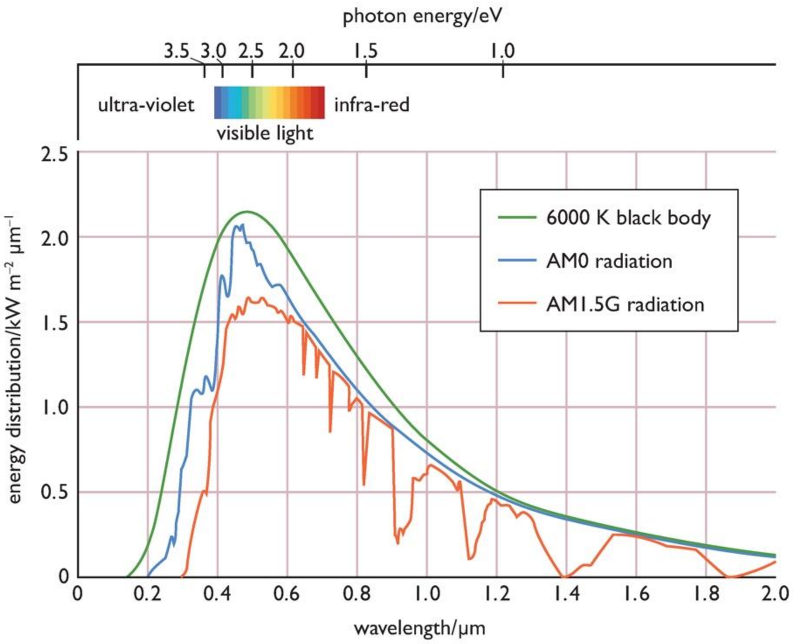
Standard Test Conditions for PV Cells



One "Full Sun" (1-sun) = 1kW/m^2

Standard Test Conditions (cont'd)

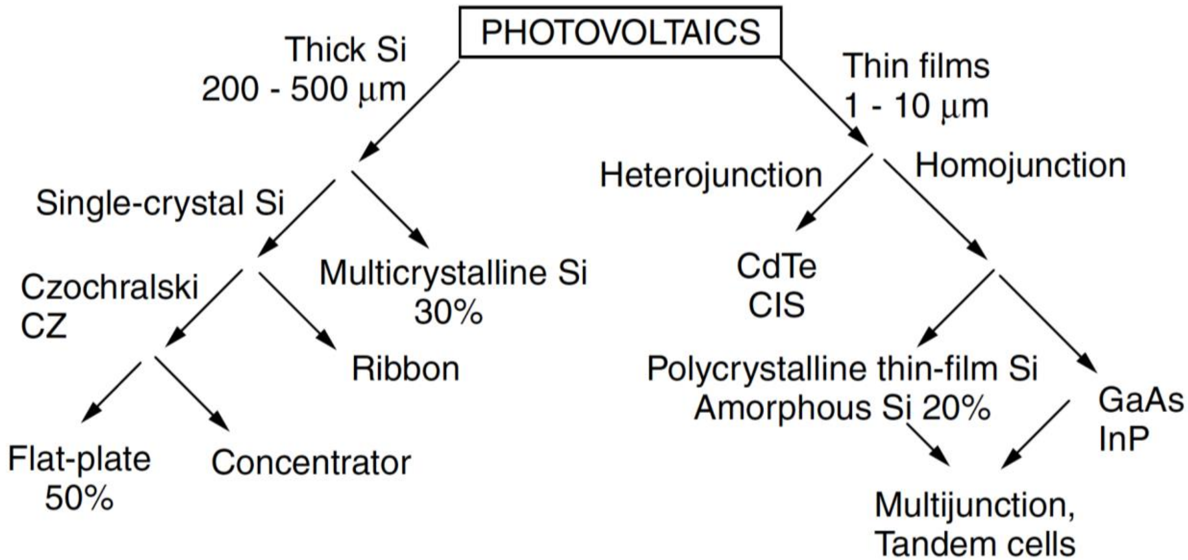


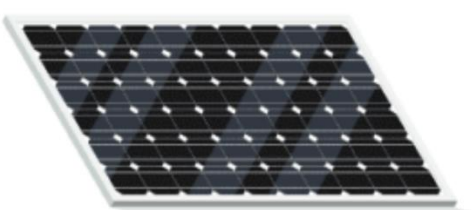


PV cell performance is dependent on solar radiation whose spectrum varies with the height of the Sun:

- AM0: Solar radiation measured in space
- AM1: Sun directly overhead
- AM1.5: Sun at 48^o angle
- 6K black body: Assuming Sun is an ideal 'black body' radiator with a surface temperature of 6000^o K.

PV Tech Classification



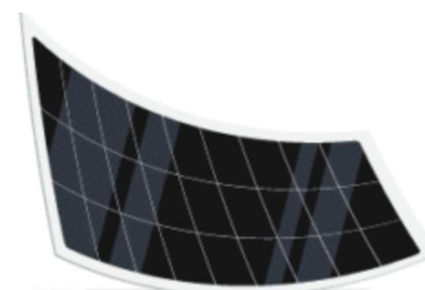


Monocrystalline

- Highest price
- Typically over **19%** efficiency
- Excellent choice when you have limited roof space

Polycrystalline

- Intermediate price
- Typically **15-17%** efficiency
- Great choice if space is not a limitation



Thin-film

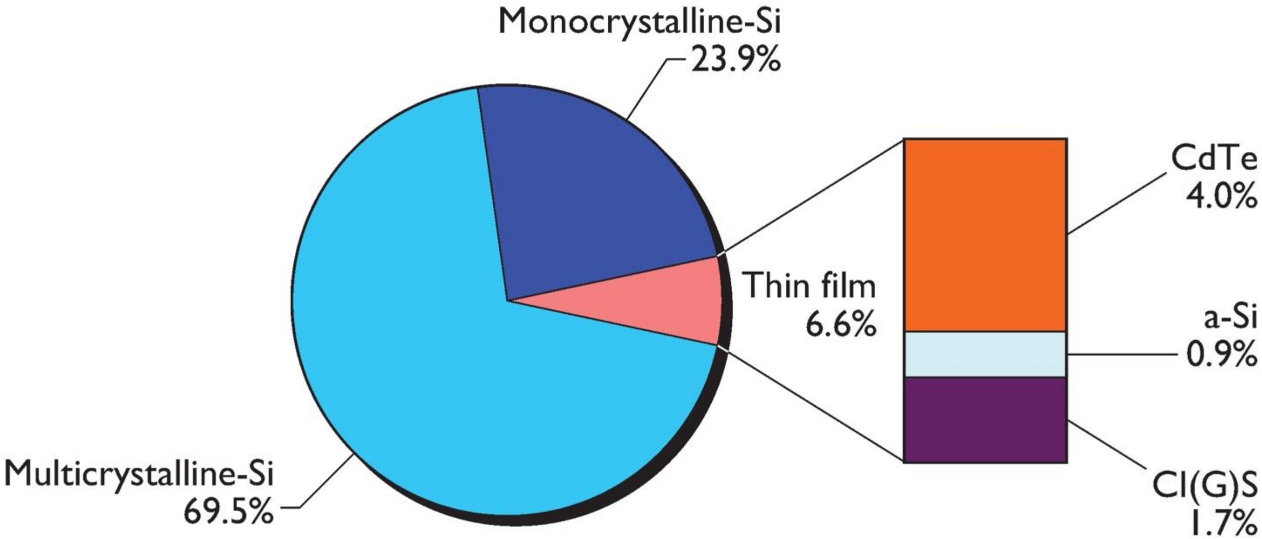
- Lowest price
- Typically below **15%** efficiency
- Best for large commercial and industrial rooftops

https://www.ecowatch.com/solar/thin-film-solar-panels

Global Production for Different PV Tech

- Crystalline silicon (c-Si), used in traditional wafer-based solar cells:
  - Monocrystalline silicon (mono-Si)
  - Multicrystalline silicon (multi-Si)

- Non-crystalline silicon, used in thin-film & other solar cell tech:
  - Amorphous silicon (a-Si): low cost but relatively low efficiency
  - Nanocrystalline silicon (nc-Si): form of porous silicon
  - Protocrystalline silicon (pc-Si): distinct phase occurring during crystal growth
  - Other non-silicon materials, such as CdTe, CIGS



Crystalline-Silicon vs Thin-film

Feature	Thin-Film PV Cells	Crystalline Silicon (c-Si) PV Cells
Materials	a-Si, CdTe, CIGS	Monocrystalline Si, Polycrystalline Si
Structure	Amorphous/microcrystalline thin layers deposited on substrates	Bulk wafers sliced from silicon ingots
Efficiency (Lab / Commercial)	~10–22% / 8–15%	~26–27% / 18–24% (mono-Si), 15–20% (poly-Si)
Cost per Watt	Lower (generally)	Higher, but decreasing with scale
Low-Light Performance	Relatively strong	Moderate
Temperature Coefficient	Better (lower performance degradation at high temperatures)	Worse (higher degradation with heat)
Weight & Flexibility	Lightweight; can be flexible (substrate-dependent)	Heavier; rigid
Manufacturing	Scalable deposition (e.g., roll-to-roll possible)	High-purity Si processing and precision wafering required
Material Availability	May involve scarce/toxic elements (e.g., Cd in CdTe)	Abundant (Si); environmentally safer
Durability / Lifetime	Shorter (~20–25 years)	Longer (~25–30+ years)
Market Share (≈2025)	<10% (primarily CdTe)	>90% (dominated by mono-Si)
Typical Applications	BIPV, flexible modules, weight/space-constrained systems	Utility-scale, rooftop, high-efficiency applications

Crystalline PV Cells

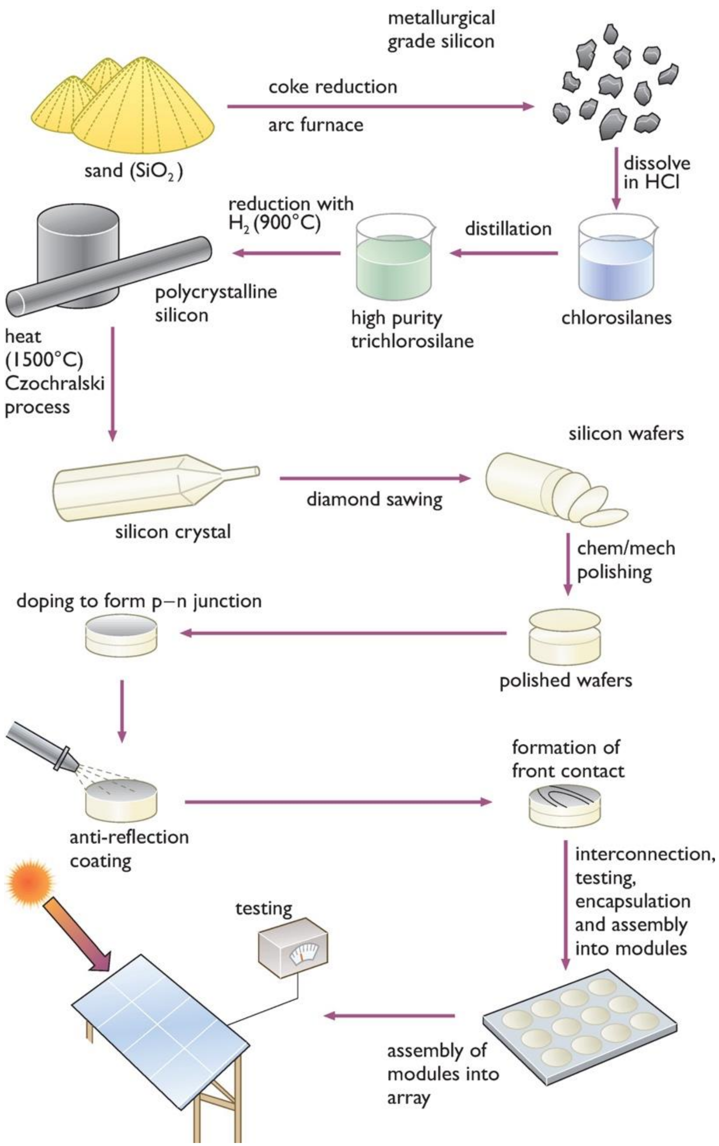
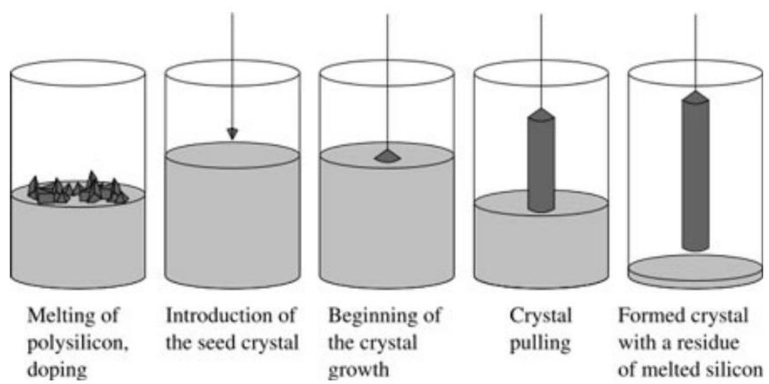


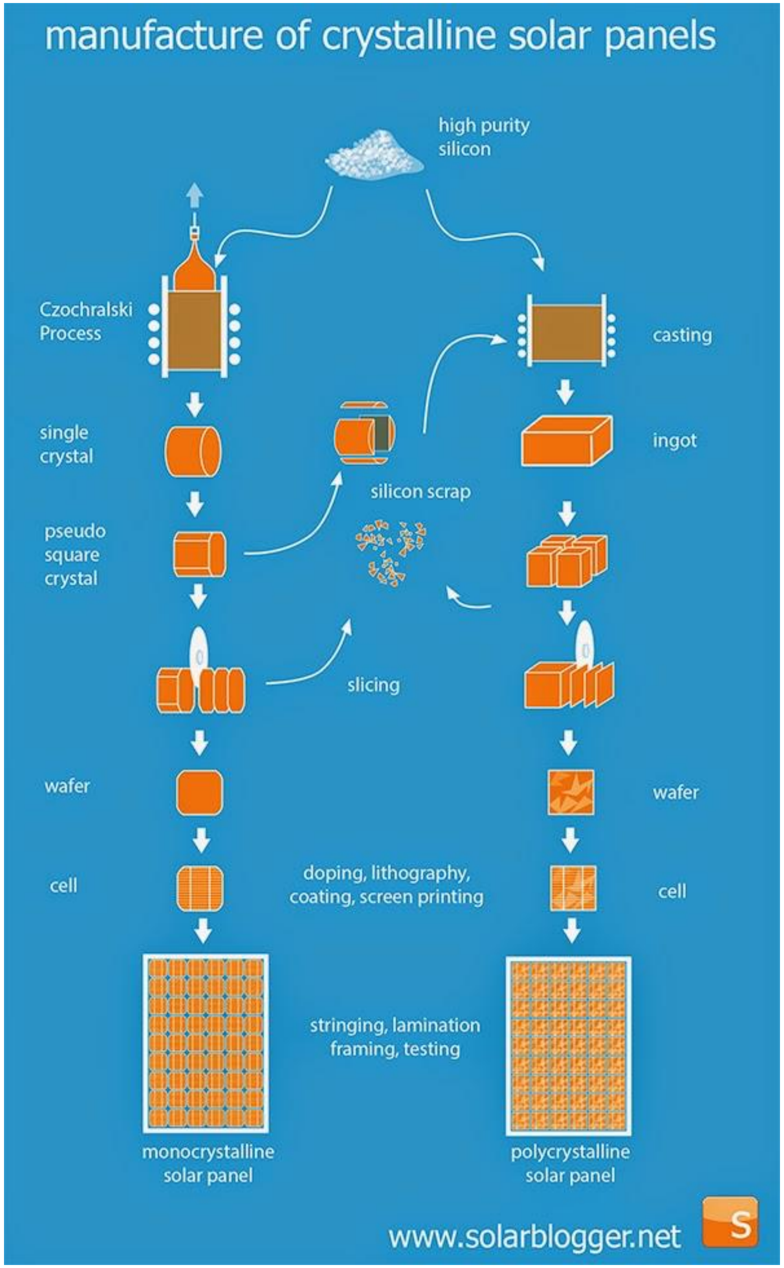
Fig: The process of monocrystalline silicon solar cell and module production

The most efficient cells are made from extremely pure monocrystalline: silicon with a single, continuous crystal lattice structure with no defects/impurities.

The **Czochralski process:** A method of crystal growth used to obtain single crystals of semiconductors (e.g. silicon, germanium and gallium arsenide), metals, salts and synthetic gemstones.



Mono vs Poly Crystalline



- **Monocrystalline** silicon is usually grown from a small seed crystal that is slowly pulled out of a molten mass of polycrystalline silicon (consists of small grains of monocrystalline).
- Polycrystalline (multicrystalline) cells are easier and cheaper to manufacture, but are less efficient.

Thin-film Silicon PV

Solar cells can also be made from very thin films of silicon, in a form known as amorphous silicon (a-Si), in which the silicon atoms are much less ordered

A-Si cells are cheaper to produce. Advantages in manufacturing process:

- ✓ operates at a much lower temperature, less energy required
- ✓ suited to continuous production
- ✓ allows quite large areas of cells to be deposited onto a wide variety of both rigid and flexible substrates, including steel, glass, and plastics.

A-Si cells are currently much less efficient than the mono/poly-crystalline silicon counterparts. They are already widely used as power sources for a variety of consumer products, where the requirement is for low cost rather than high efficiency.

Other Thin-film Silicon PV Tech

Based on compound semiconductors:

- Copper indium gallium diselenide (CIGS)
- Cadmium telluride (CdTe)

Categories	Technology	η (%)	V_{OC} (V)	I_{SC} (A)	W/m²	t (μm)
Thin film solar cells	a-Si	11.1	6.3	0.0089	33	1
	CdTe	16.5	0.86	0.029	–	5
	CIGS	20.5	–	–	–	

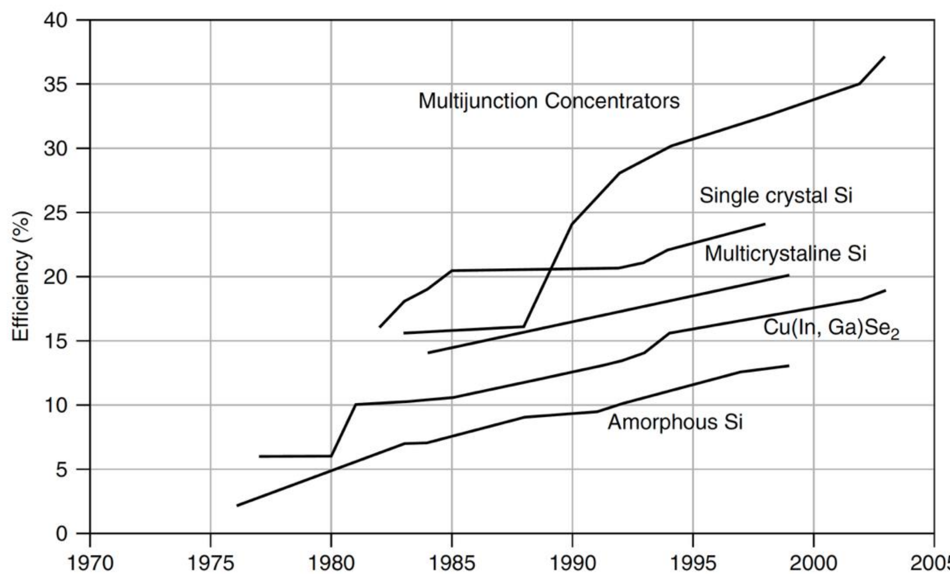


Figure 8.1 Best laboratory PV cell efficiencies for various technologies. (From National Center for Photovoltaics, www.nrel.gov/ncpv 2003).

Copper indium gallium (di)selenide (CIGS)

- CIGS is a I-III-VI_2 semiconductor material composed of copper, indium, gallium, and selenium. The material is a solid solution of copper indium selenide ("CIS") and copper gallium selenide.
- CIGS has a chemical formula of CuIn(1-x)Ga(x)Se2 where the value of x can vary from 0 (pure copper indium selenide) to 1 (pure copper gallium selenide).
- CIGS cells have the highest laboratory efficiencies of all thin-film devices, around 17%; modules over 10% are *commercially* available.
- CIGS has a bandgap varying continuously with x from about 1.0 eV (copper indium selenide) to about 1.7 eV (copper gallium selenide).

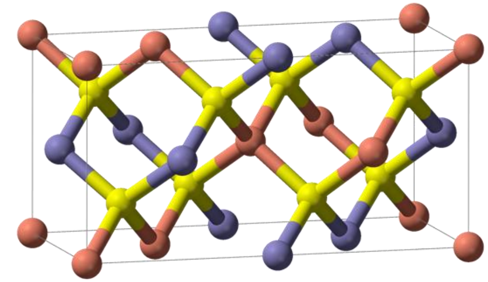


Fig a: CIGS unit cell. Red = Cu, yellow = Se, blue = In/Ga.

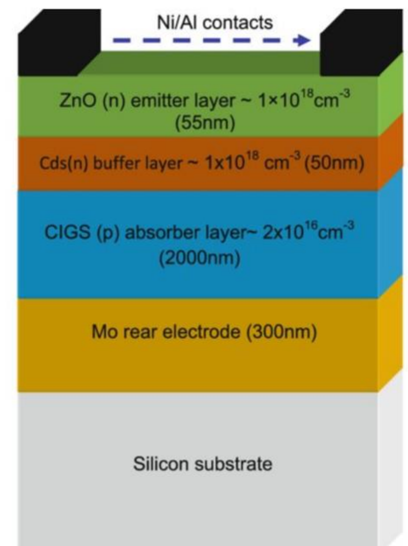
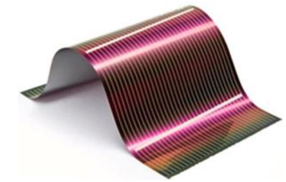


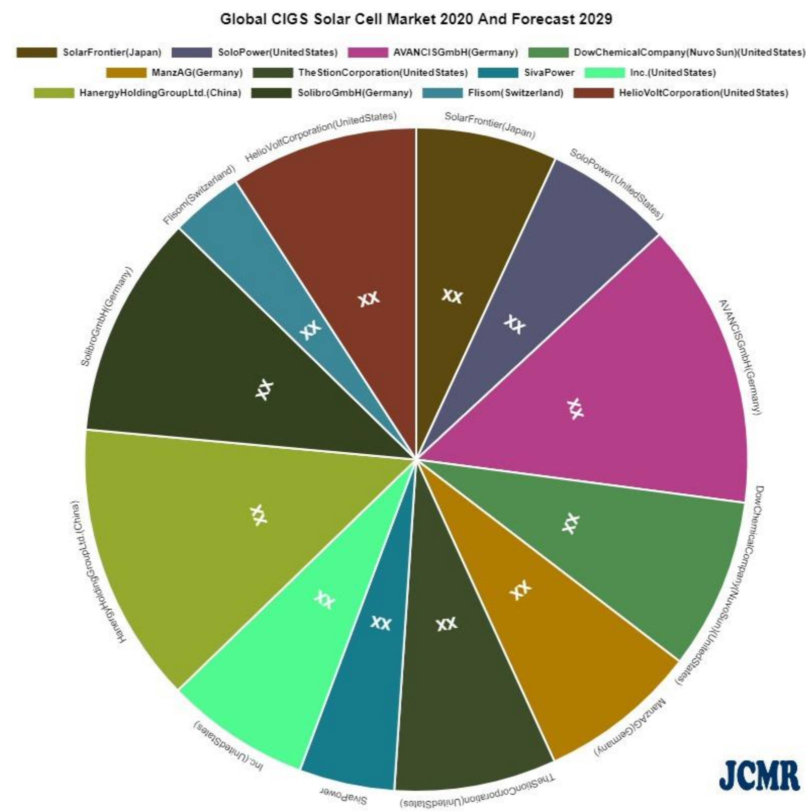
Fig b: Structure of the CIGS solar cell. ZnO = Zinc oxide, CdS = Cadmium sulfide, Mo = Molybdenum.

See more in https://www.hisour.com/copper-indium-gallium-selenide-solar-cells-39668/

Benefits of CIGS Solar Cells

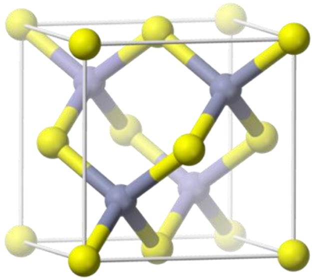


- High absorption: This direct-bandgap material can absorb a significant portion of the solar spectrum, enabling it to achieve the highest efficiency of any thin-film technology.
- Tandem design: A tunable bandgap allows the tandem CIGS devices.
- Protective buffer layer: The grain boundaries form an inherent buffer layer, preventing surface recombination and allowing for films with grain sizes of less than 1 micrometer to be used in device fabrication.

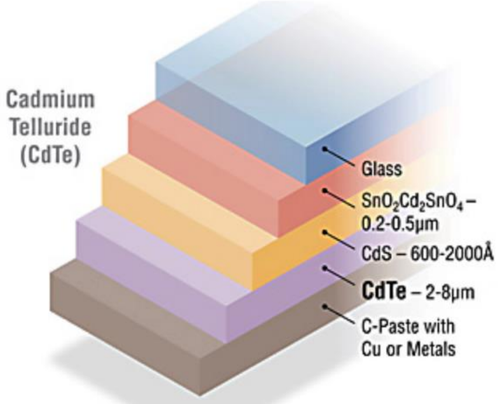


Source: https://www.energy.gov/eere/solar/copper-indium-gallium-diselenide

Cadmium Telluride (CdTe)



- CdTe-based PV is a thin-film technology: active layers are just a few microns thick (1/10 the diameter of a human hair).
- Transparent conducting oxide (TCO) layers (SnO_2/Cd_2SnO_4) are transparent to visible light and highly conductive to transport current efficiently.
- Intermediate layers such as CdS help in both the growth and electrical properties between the TCO and CdTe that acts as the primary photoconversion layer and absorbs most visible light.
- All three layers form an electric field that converts light absorbed in the CdTe layer into I & V.
- CdTe: relatively simple and inexpensive process. But cadmium is a highly toxic substance. Precautions need during manufacture/use/recycling.



See more in https://www.hisour.com/copper-indium-gallium-selenide-solar-cells-39668/

Other PV Materials - Gallium Arsenide Cells

Gallium arsenide (GaAs):

• structure similar to silicon, consist of gallium and arsenic atoms

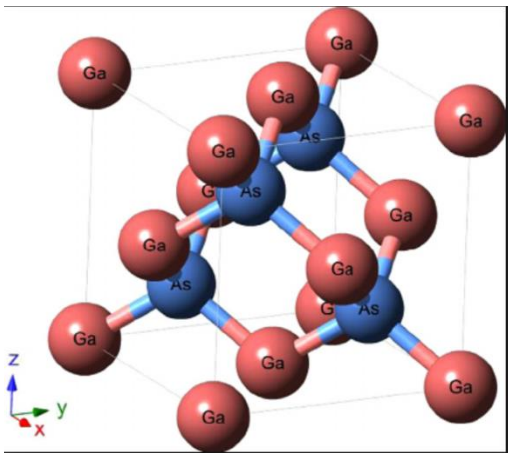
• more efficient than monocrystalline cells

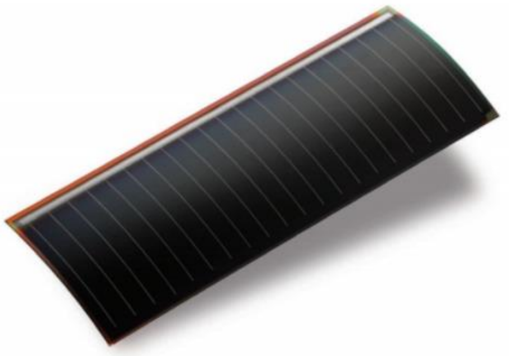
• high light absorption coefficient (need only a thin layer)

• can operate at relatively high temperatures without substantial reduction in efficiency

• very expensive: i) gallium and arsenic not abundant; and ii) partly production process is not well developed

• often used when very high efficiency is required, regardless of cost – e.g., space PV applications.





Multi-junction PV Cells

Multi-junction solar cells (MJ): multiple p–n junctions made of different semiconductor materials. Each material's p-n junction will produce electric current in response to different wavelengths of light

Prize-winning Tokai Challenger solar car:

• An array of more than 2000 Sharp triple-junction PV cells, with a peak power output of 1.8 kW and an efficiency of 30%, was used to power the electric motors

• It covered 2998 km in 29 hours 49 minutes at an average speed of 100.54 km/hour (62mph).